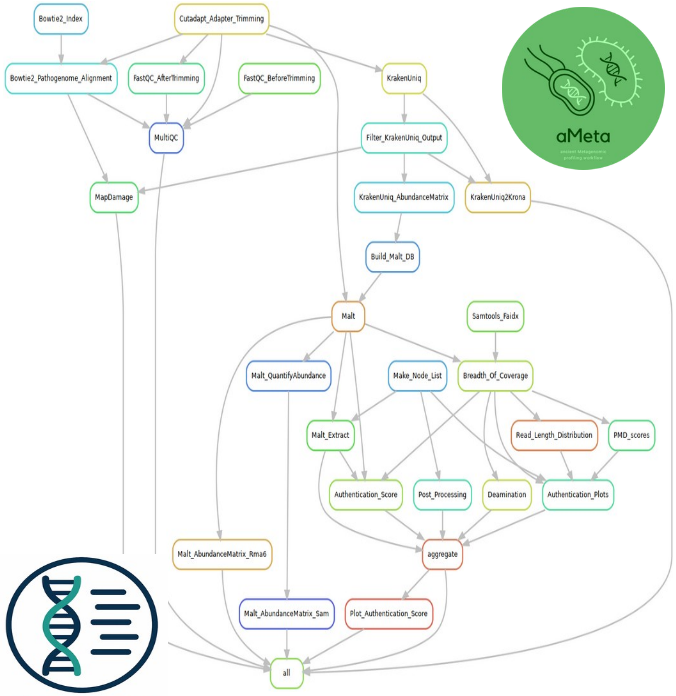
Computational challenges in ancient metagenomics

Nikolay Oskolkov, Metabolic Research Group leader, NIRI, Latvia, and Lund University, Sweden

Physalia Ancient Metagenomics course, online, 2-4 of June 2026



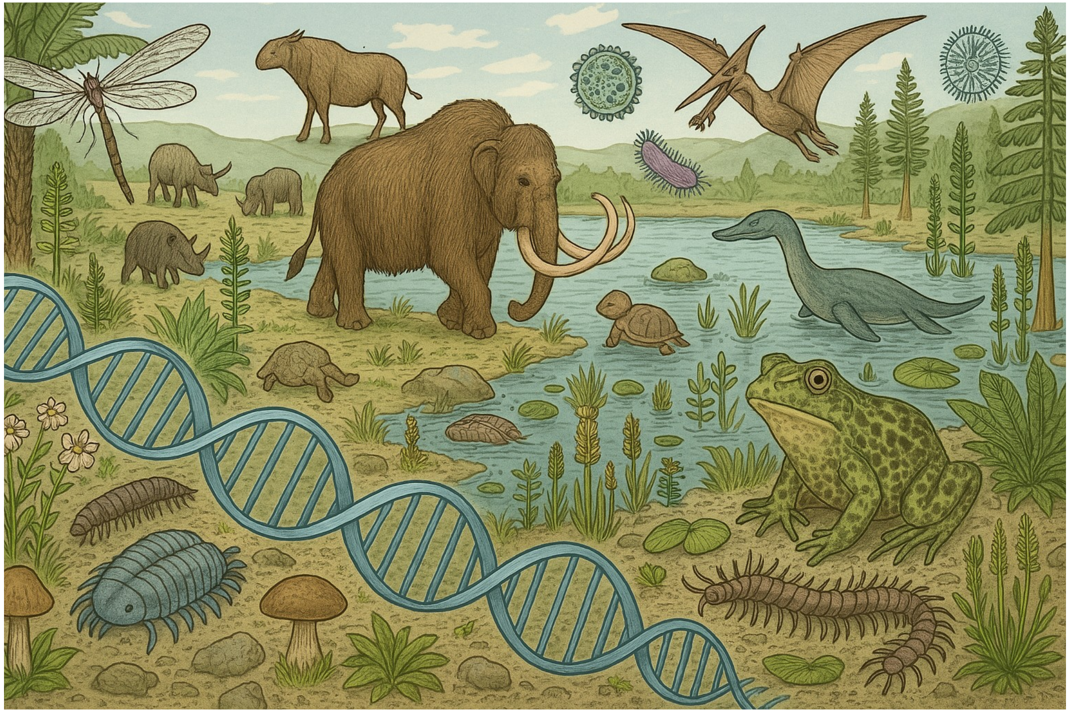


Image generated by ChatGPT (OpenAI), April 2025

@NikolayOskolkov

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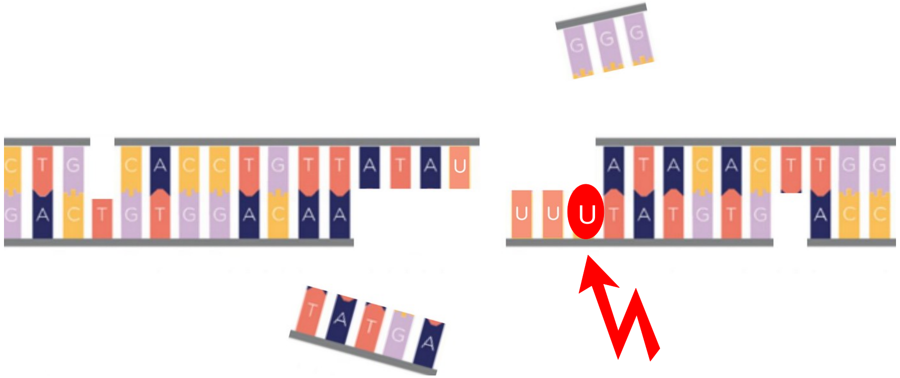
Personal homepage:
https://nikolay-oskolkov.com

Ancient DNA from archaeological samples

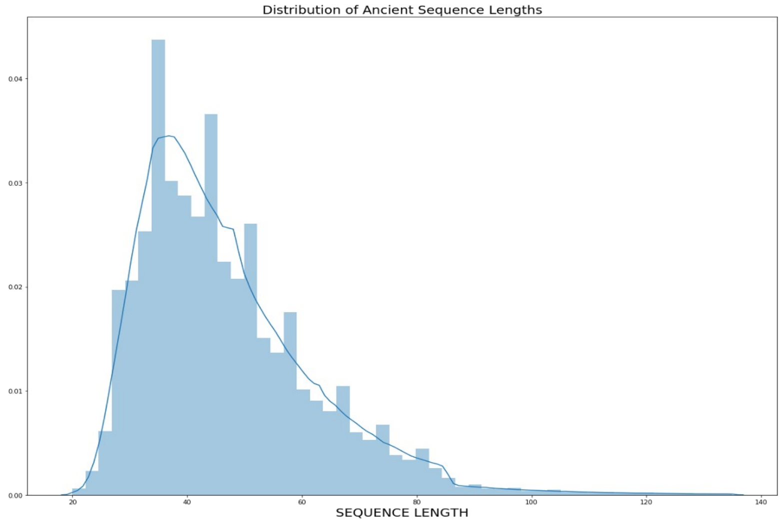

TTCAGGAAAACCCGATAAAAATTCTTTATTGGGGGAGGGGCTCAAACAAGAAAATAATCAACAAGTGGTGTCCAGAGTGGAGCCAGGG
CCTCCTGGGGACAGCAAGACTGCCTGGGGAGCGGGAAGCAGCTCCCCCGTCTCTGGGGGAGGCGTGGCTGGAGGGGAGGCTGGACCA
CAGGAGGGCAGCGCCCTGGGCAACCCTATGTAGATGAAGCTGCCGGAGAGGATCAAAGAACCAGACAGGAGGAAAGAGGCGGTGAAG
TCTCCTGTCTCATCCCTTAGGAAGCCTGAGGAGATGGGTAAGGGCATTTAGAAGCCTCGAACCCCAGGGGCAGAGGATAGTTGTAGGCA

Peculiarity: very few samples with little amounts of DNA, hence hard for statistical analysis

Ancient DNA (aDNA) degradation

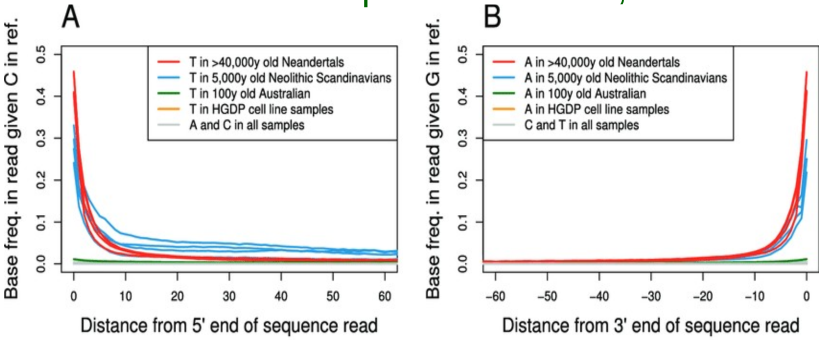


DNA fragmentation



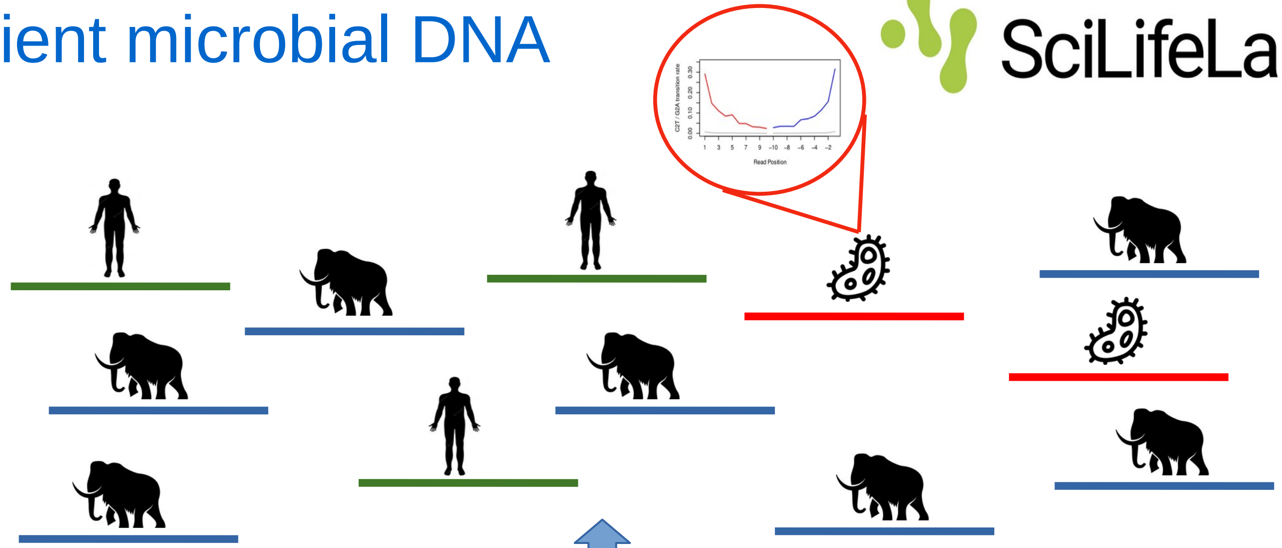
Damage / deamination pattern

Deamination process: C → T, G → A

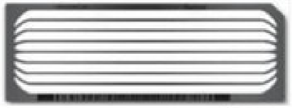


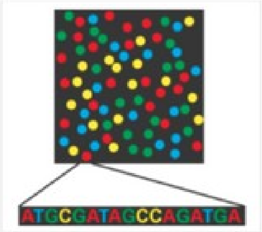
Skoglund et al. 2014

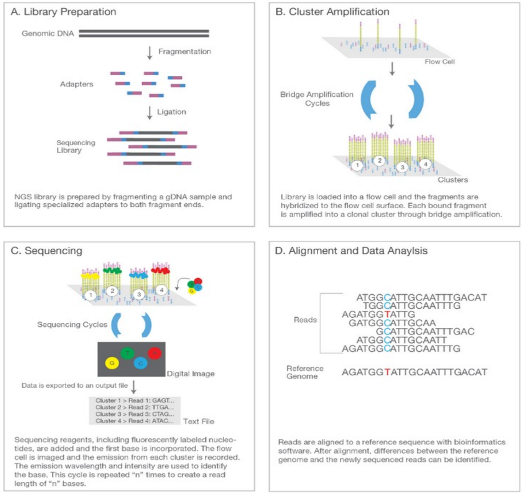
Ancient microbial DNA



Sequence







Ancient microbial pathogens

Proc. Natl. Acad. Sci. USA
Vol. 95, pp. 12637–12640, October 1998
Microbiology

Detection of 400-year-old *Yersinia pestis* DNA in human dental pulp: An approach to the diagnosis of ancient septicemia

(ancient DNA/paleomicrobiology/*pla* gene/*rpoB* gene)

MICHEL DRANCOURT*, GÉRARD ABOUDHARAM*, MICHEL SIGNOLI†, OLIVIER DUTOUR†, AND DIDIER RAOULT*‡

*Unité des Rickettsies, Centre National de la Recherche Scientifique UPRES-A 6020, Université de la Méditerranée, and †Laboratoire d'Anthropologie biologique, Unit, Mixte de Recherche, Centre National de la Recherche Scientifique-Université de la Méditerranée Association de Fouilles Archeologiques Nationales, Faculté de Médecine, 27, Boulevard Jean Moulin, 13385 Marseille Cédex 05, France

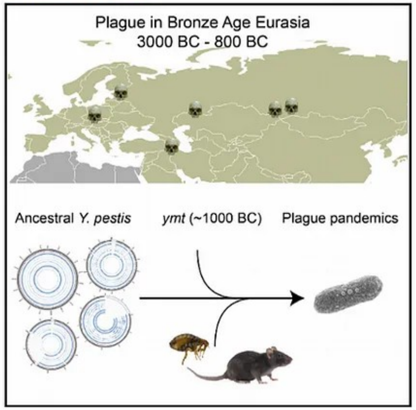
Edited by Joshua Lederberg, The Rockefeller University, New York, NY, and approved August 10, 1998 (received for review May 12, 1998)

Article

Cell

Early Divergent Strains of *Yersinia pestis* in Eurasia 5,000 Years Ago

Graphical Abstract



Authors

Simon Rasmussen, Morten Erik Allentoft, Kasper Nielsen, ..., Rasmus Nielsen, Kristian Kristiansen, Eske Willerslev

Correspondence

ewillerslev@snm.ku.dk

In Brief

The plague-causing bacteria *Yersinia pestis* infected humans in Bronze Age Eurasia, three millennia earlier than any historical records of plague, but only acquired the genetic changes making it a highly virulent, flea-borne bubonic strain ~3,000 years ago.

Science

RESEARCH ARTICLE

Diverse variola virus (smallpox) strains were widespread in northern Europe in the Viking Age

BARBARA MÜHLEMANN, LASSE VINNER, ASHOT MARGARYAN, HELENE WILHELMSON, CONSTANZA DE LA FUENTE CASTRO, MORTEN E. ALLENTOFT, PETER DE BARROS DAMGAARD, ANDERS JOHANNES HANSEN, SOFIE HOLTSMARK NIELSEN, [...], AND MARTIN SIKORA +19 authors

SCIENCE · 24 Jul 2020 · Vol 369, Issue 6502 · DOI: 10.1126/science.aaw8977

PNAS

Genomic blueprint of a relapsing fever pathogen in 15th century Scandinavia

Meriam Guellil[a,1], Oliver Kersten[a], Amine Namouchi[a], Egil L. Bauer[b], Michael Derrick[b], Anne Ø. Jensen[b], Nils C. Stenseth[a,1], and Barbara Bramanti[a,c,1]

Louse-borne relapsing fever (LBRF) is known to have killed millions of people over the course of European history and remains a major cause of mortality in parts of the world. Its pathogen, *Borrelia recurrentis*, shares a common vector with global killers such as typhus and plague and is known for its involvement in devastating historical epidemics such as the Irish potato famine. Here, we describe a European and historical genome of *B. recurrentis*, recovered from a 15th century skeleton from Oslo. Our distinct European lineage has a discrete genomic makeup, displaying an ancestral *oppA-1* gene and gene loss in antigenic variation sites. Our results illustrate the potential of ancient DNA research to elucidate dynamics of reductive evolution in a specialized human pathogen and to uncover aspects of human health usually invisible to the archaeological record.

MICROBIAL GENOMICS

REVIEWS

Ancient pathogen genomics as an emerging tool for infectious disease research

Maria A. Spyrou*, Kirsten I. Bos, Alexander Herbig and Johannes Krause*

Abstract | Over the past decade, a genomics revolution, made possible through the development of high-throughput sequencing, has triggered considerable progress in the study of ancient DNA, enabling complete genomes of past organisms to be reconstructed. A newly established branch of this field, ancient pathogen genomics, affords an in-depth view of microbial evolution by providing a molecular fossil record for a number of human-associated pathogens. Recent accomplishments include the confident identification of causative agents from past pandemics, the discovery of microbial lineages that are now extinct, the extrapolation of past emergence events on a chronological scale and the characterization of long-term evolutionary history of microorganisms that remain relevant to public health today. In this Review, we discuss methodological advancements, persistent challenges and novel revelations gained through the study of ancient pathogen genomes.

nature

Letter | Published: 09 May 2018

Ancient hepatitis B viruses from the Bronze Age to the Medieval period

Barbara Mühlemann, Terry C. Jones, Peter de Barros Damgaard, Morten E. Allentoft, Irina Shevnina, Andrey Logvin, Emma Usmanova, Irina P. Panyushkina, Bazartseren Boldgiv, Tsevel Bazartseren, Kadicha Tashbaeva, Victor Merz, Nina Lau, Václav Smrčka, Dmitry Voyakin, Egor Kitov, Andrey Epimakhov, Dalia Pokutta, Magdolna Vicze, T. Douglas Price, Vyacheslav Moiseyev, Anders J. Hansen, Ludovic Orlando, Simon Rasmussen, … Eske Willerslev + Show authors

Nature 557, 418–423 (2018)

BMC Part of Springer Nature

Genome Biology

Research | Open access | Published: 10 August 2020

A seventeenth-century *Mycobacterium tuberculosis* genome supports a Neolithic emergence of the *Mycobacterium tuberculosis* complex

Susanna Sabin, Alexander Herbig, Åshild J. Vågene, Torbjörn Ahlström, Gracijela Bozovic, Caroline Arcini, Denise Kühnert & Kirsten I. Bos

Genome Biology 21, Article number: 201 (2020)

Science

REPORT

The 5300-year-old *Helicobacter pylori* genome of the Iceman

FRANK MAIXNER, BEN KRAUSE-KYORA, DMITRIJ TURAEV, ALEXANDER HERBIG, MICHAEL R. HOOPMANN, JANICE L. HALLOWS, ULRIKE KUSEBAUCH, EDUARD EGARTER VIGL, PETER MALFERTHEINER, [...] AND ALBERT ZINK +13 authors

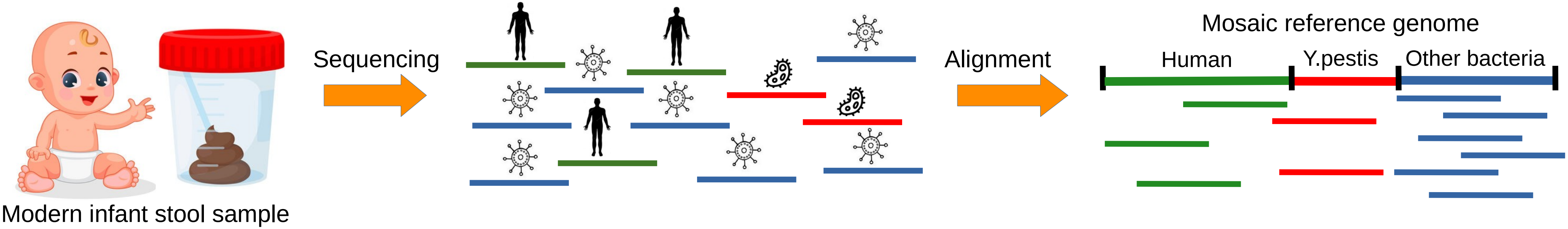
SCIENCE · 8 Jan 2016 · Vol 351, Issue 6269 · pp. 162-165 · DOI: 10.1126/science.aad2545

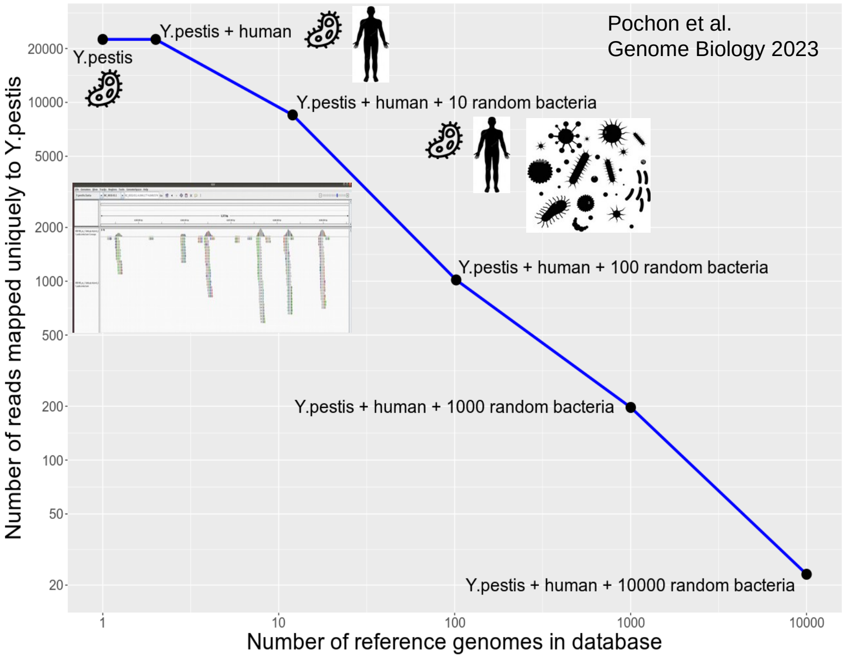
Stomach ache for a European mummy

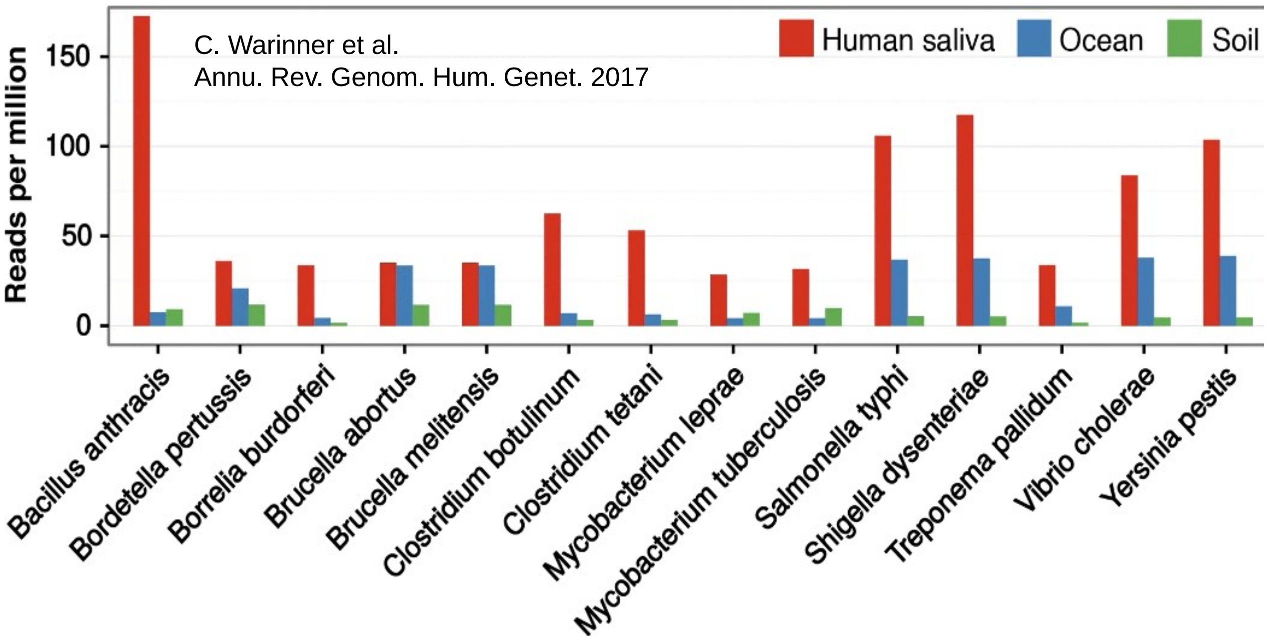
Five thousand years ago in the European Alps, a man was shot by an arrow, then clubbed to death. His body was subsequently mummified by ice until glacier retreat exhumed him in 1991. Subsequently, this ancient corpse has provided a trove of intriguing information about copper-age Europeans. Now, Maixner *et al.* have identified the human pathogen *Helicobacter pylori* within the mummy's stomach contents. The strain the "Iceman" hosted appears to most closely resemble pathogenic Asian strains found today in Central and Southern Asia.

Computational challenges in ancient <u>microbial</u> metagenomics

False-positives in microbial metagenomics







False-positives in pathogen metagenomics

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Plague, anthrax and cheese? Scientists map bacteria on New York subway

- Research strongly downplays threat as most of the bacteria is harmless
- More than 1,000 samples were run through a DNA sequencer

Alan Yuhas *in New York*
Sat 7 Feb 2015 18.46 CET
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Don't Worry, There's Probably No Bubonic Plague on NYC Subways

2 MINUTE READ

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LETTER • Volume 1, Issue 1, P4-5, July 29, 2015 • *Open Archive* | Download Full Issue

Lack of Evidence for Plague or Anthrax on the New York City Subway

Joel Ackelsberg[1] • Jennifer Rakeman[1] • Scott Hughes[1] • Jeannine Petersen[2] • Paul Mead[2] • Martin Schriefer[2] • Luke Kingry[2] • Alex Hoffmaster[3] • Jay E. Gee[3] Show less

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Show Outline

Main Text

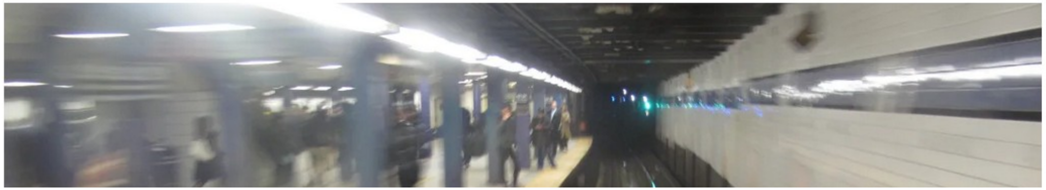
In their highly publicized report on the metagenomics of the NYC subway, Afshinnekoo and colleagues display an unfamiliarity with the genetics, microbiology, ecology, and epidemiology of some of the organisms they claim to have identified (Afshinnekoo et al., 2015).

Yersinia pestis, the cause of plague, was first introduced into North America circa 1900 through port cities along the Pacific and Gulf coasts. The organism spread into native rodent populations and became established in the arid western United States. Although a few cases occurred initially in New Orleans, LA, and Pensacola, FL, none were observed along the Atlantic coast, and the organism quickly died out in all areas east of Texas (Kugeler et al., 2015). Rodents, cats, and humans are all exquisitely susceptible to infection with *Y. pestis*, yet naturally occurring infection has never been observed within 1,000 miles of NYC. A plague outbreak in NYC's urban rat population, let alone sporadic human disease, would not go unnoticed. The authors' suggestion that humans and plague bacilli have "interacted (and potentially evolved)" in NYC is unfounded and without scientific merit.

CNN US | Crime + Justice | Watch

New York subway germ study derailed

By Lorenzo Ferrigno, CNN
2 minute read · Published 1:17 PM EDT, Fri August 14, 2015



Now the team of scientists, who spent 17 months in New York's sprawling subway system collecting microorganisms, say there is "minimal coverage to the backbone genome of these organisms, and there is no strong evidence to suggest these organisms are in fact present."

Mixing up human and microbial DNA

nature

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nature > articles > article

Article | Published: 11 March 2020

RETRACTED ARTICLE: Microbiome analyses of blood and tissues suggest cancer diagnostic approach

Gregory D. Poore, Evguenia Kopylova, Qiyun Zhu, Carolina Carpenter, Serena Fraraccio, Stephen Wandro, Tomasz Kosciolek, Stefan Janssen, Jessica Metcalf, Se Jin Song, Jad Kanbar, Sandrine Miller-Montgomery, Robert Heaton, Rana Mckay, Sandip Pravin Patel, Austin D. Swafford & Rob Knight

Nature 579, 567–574 (2020) | Cite this article

107k Accesses | 675 Citations | 979 Altmetric | Metrics

This article was retracted on 26 June 2024

This article has been updated

Abstract

Systematic characterization of the cancer microbiome provides the opportunity to develop techniques that exploit non-human, microorganism-derived molecules in the diagnosis of a major human disease. Following recent demonstrations that some types of cancer show substantial microbial contributions[1,2,3,4,5,6,7,8,9,10], we re-examined whole-genome and whole-transcriptome sequencing studies in The Cancer Genome Atlas[11] (TCGA) of 33 types of cancer from treatment-naive patients (a total of 18,116 samples) for microbial reads, and found unique microbial signatures in tissue and blood within and between most major types of cancer. These TCGA blood signatures remained predictive when applied to patients with stage Ia–IIc cancer and cancers lacking any genomic alterations currently measured on two commercial-grade cell-free tumour DNA platforms, despite the use of very stringent decontamination analyses that discarded up to 92.3% of total sequence data. In addition, we could discriminate among samples from healthy, cancer-free individuals ($n = 69$) and those from patients with multiple types of cancer (prostate, lung, and melanoma; 100 samples in total) solely using plasma-derived, cell-free microbial nucleic acids. This potential microbiome-based oncology diagnostic tool warrants further exploration

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Associated content

RETRACTED ARTICLE: AI finds microbial signatures in tumours and blood across cancer types

Nadim J. Ajami & Jennifer A. Wargo

Nature | **News & Views** | 11 Mar 2020

Sections | Figures | References

'All authors agree' to retraction of Nature article linking microbial DNA to cancer

A 2020 paper that claimed to find a link between microbial genomes in tissue and cancer has been retracted following an analysis that called the results into question.

The paper, "Microbiome analyses of blood and tissues suggest cancer diagnostic approach," was published in March 2020 and has been cited 610 times, according to Clarivate's Web of Science. It was retracted June 26. The study was also key to the formation of biotech start-up Micronoma, which did not immediately respond to our request for comment.

Rob Knight, corresponding author and researcher at the University of California San Diego, also did not immediately respond to our request for comment.

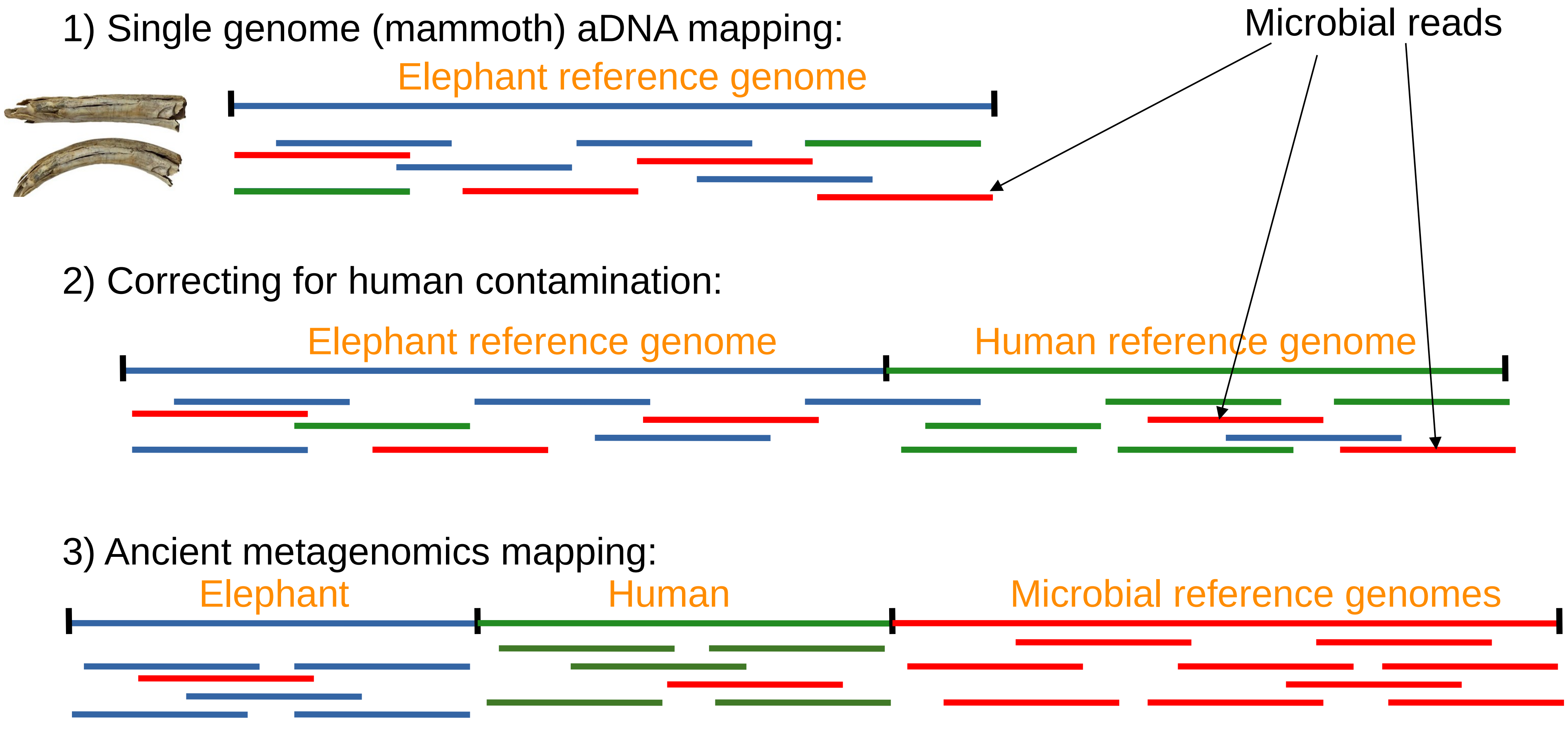
In October 2023, *mBio*, a journal from the American Society for Microbiology, published "Major data analysis errors invalidate cancer microbiome findings." The paper pointed out several major flaws in the the earlier article by Knight's group.

aMeta goal: minimize amount of false-positives

aMeta key principles

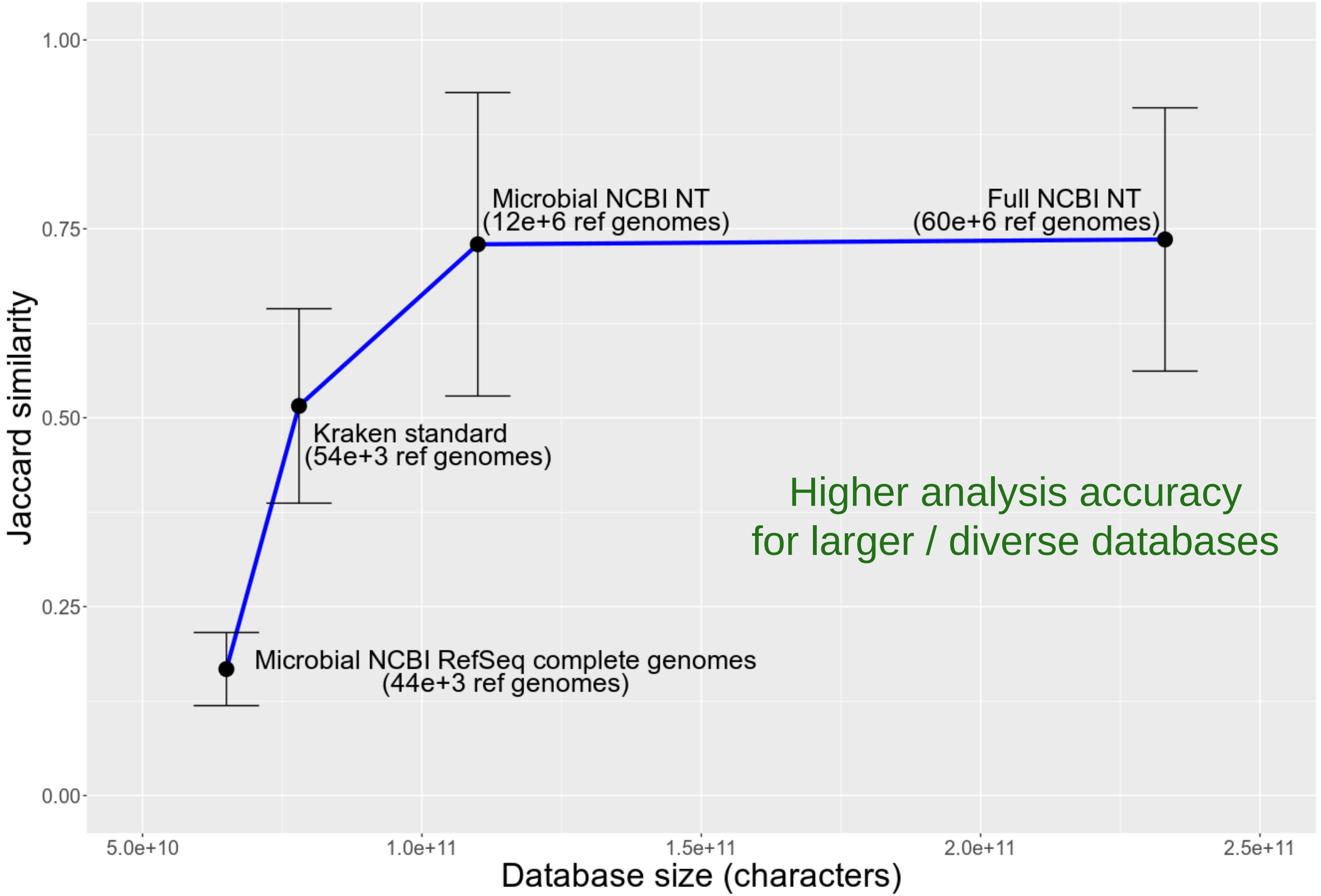
1) unbiased (ultra-)competative mapping
2) large / diverse reference databases
3) controlling evenness of coverage

Why you should do competitive mapping



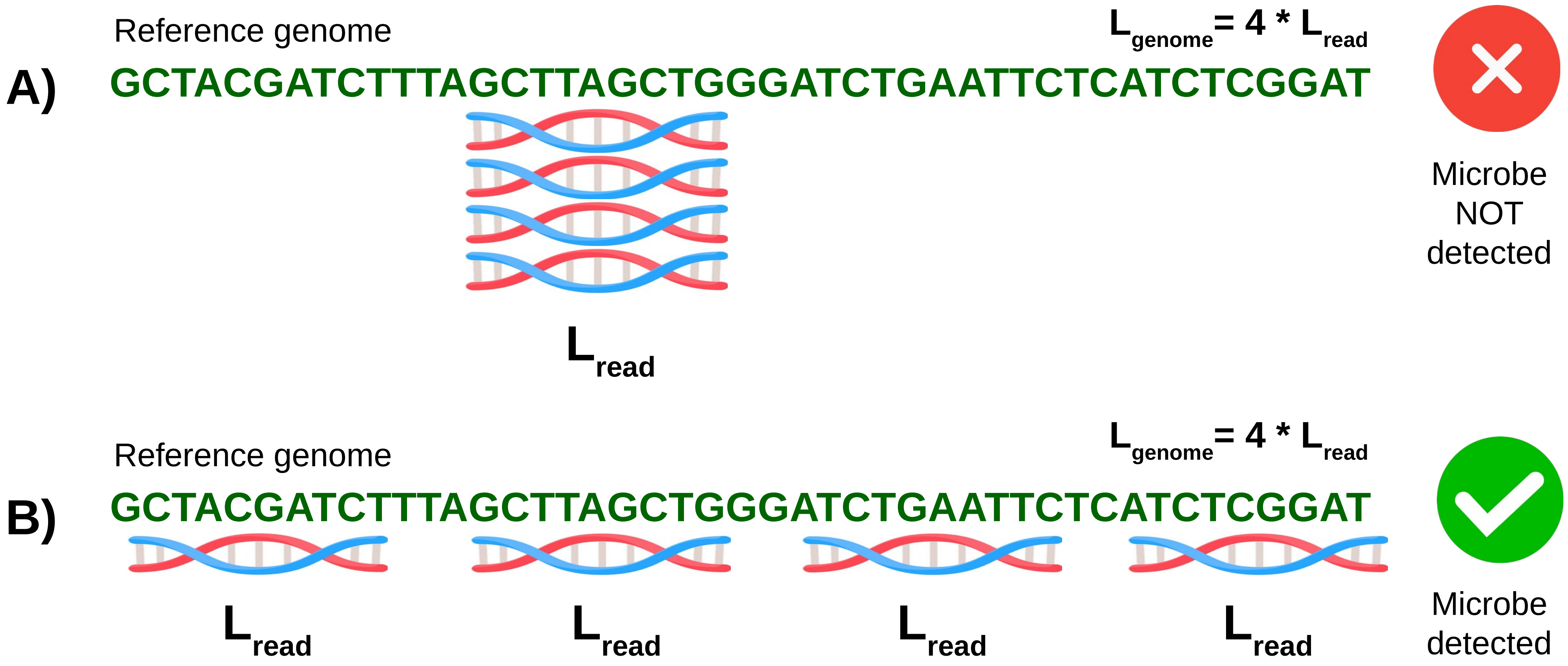
Competitive mapping is absolutely central for Ancient Metagenomics!

Why you should use large databases



Pochon et al.
Genome Biology 2023

Why you should use evenness of coverage



Both A) and B) have identical coverage:

Coverage = $(N_{reads} * L_{read}) / L_{genome} = (4 * L) / (4 * L) = 1X$

Why to develop aMeta if there is HOPS?

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Genome Biology

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Method | Open Access | Published: 16 December 2019

HOPS: automated detection and authentication of pathogen DNA in archaeological remains

Ron Hübler, Felix M. Key, Christina Warinner, Kirsten I. Bos, Johannes Krause & Alexander Herbig

Genome Biology **20**, Article number: 280 (2019) | Cite this article

5164 Accesses | **35** Citations | **26** Altmetric | Metrics

Abstract

High-throughput DNA sequencing enables large-scale metagenomic analyses of complex biological systems. Such analyses are not restricted to present-day samples and can also be applied to molecular data from archaeological remains. Investigations of ancient microbes can provide valuable information on past bacterial commensals and pathogens, but their molecular detection remains a challenge. Here, we present HOPS (Heuristic Operations for Pathogen Screening), an automated bacterial screening pipeline for ancient DNA sequences that provides detailed information on species identification and authenticity. HOPS is a versatile tool for high-throughput screening of DNA from archaeological material to identify candidates for genome-level analyses.

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Collection

Microbiome Biology

Sections   Figures   References

Abstract
Background
Results
Discussion
Conclusions
Methods
Availability of data and materials
References
Acknowledgements

H O P S

Nice design:
1) who is there?
2) how ancient?

SEVERE LIMITATIONS:

1) HOPS / MALT is very resource demanding

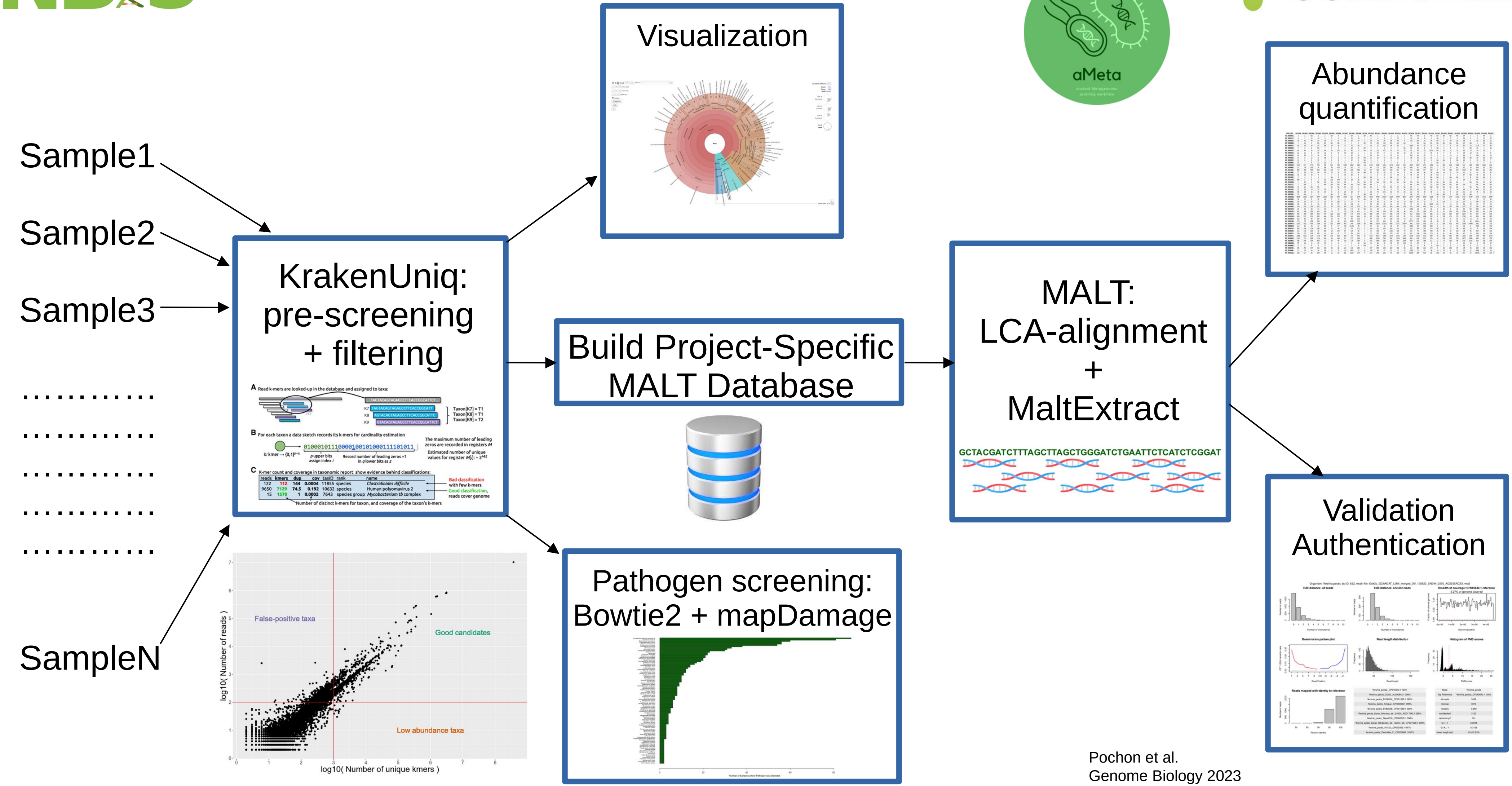
2) HOPS / MALT only feasible with limited size databases

3) HOPS does not use evenness and breadth of coverage

aMeta
ancient Metagenomic profiling workflow

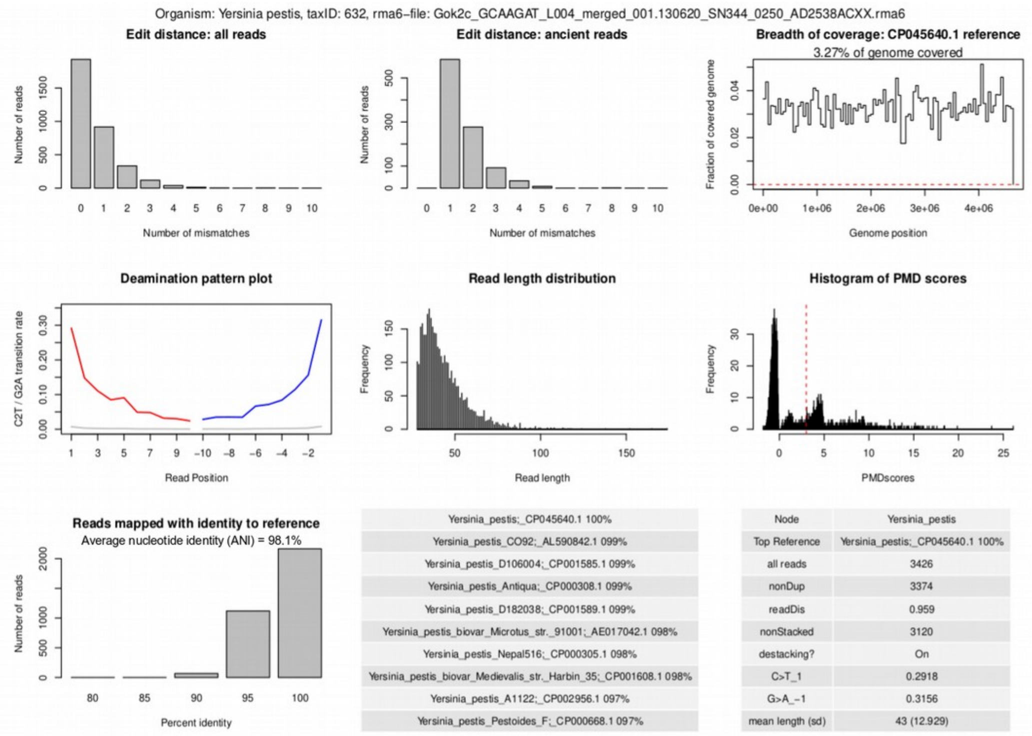
aMeta:
a way to overcome HOPS limitations

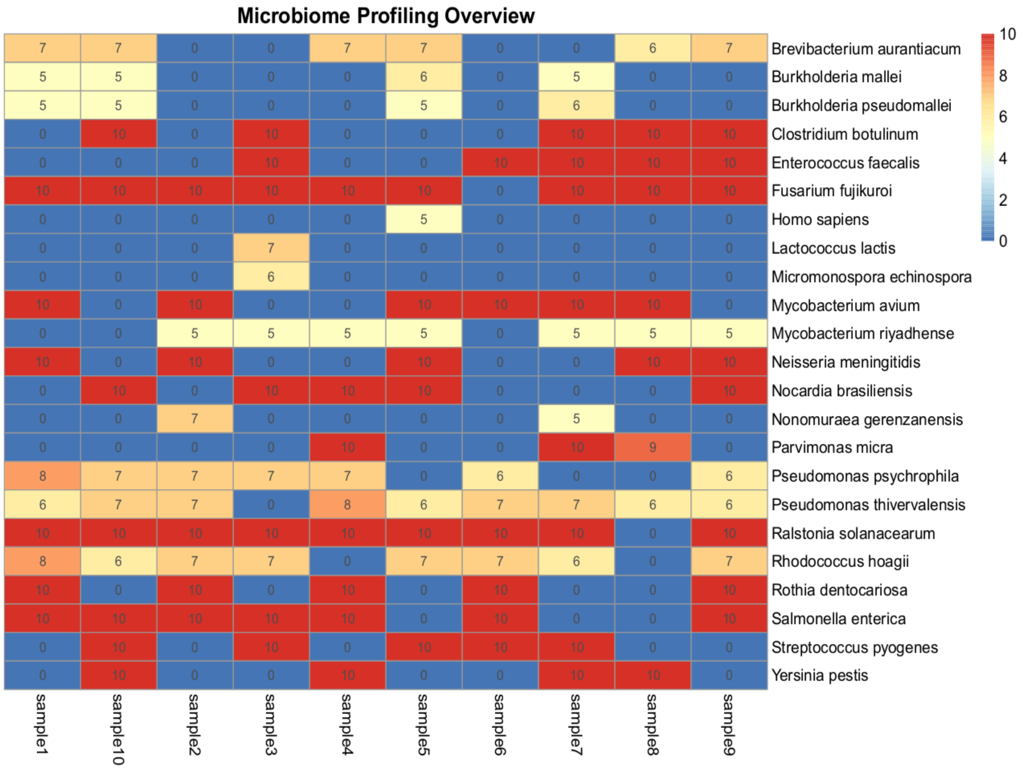
aMeta Workflow Overview



Pochon et al.
Genome Biology 2023

aMeta validation and authentication output





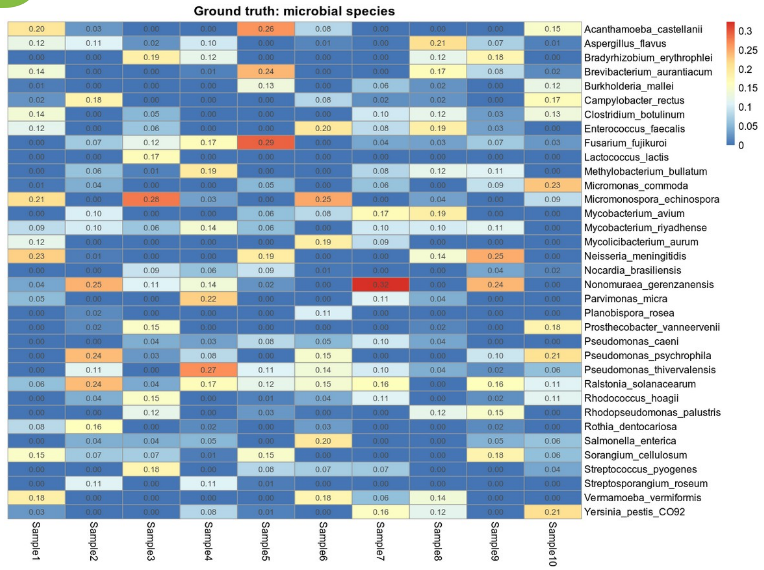
Eight quality metrics for assessing presence of microbial organisms and their ancient status

Quantified authentication and validation quality metrics as scores

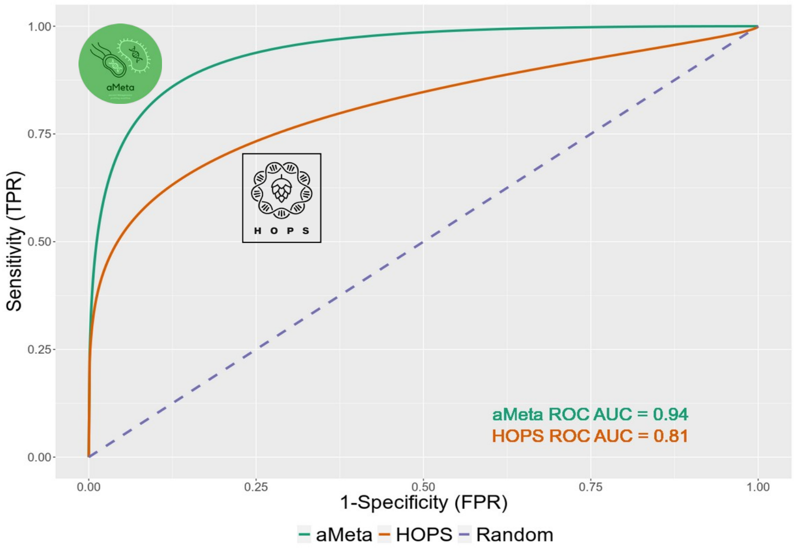
Pochon et al.
Genome Biology 2023

aMeta vs HOPS benchmarking

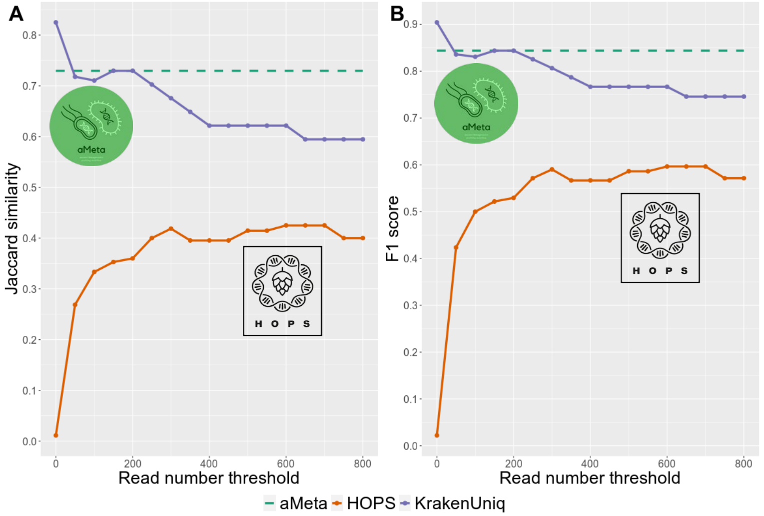
Simulated ground truth



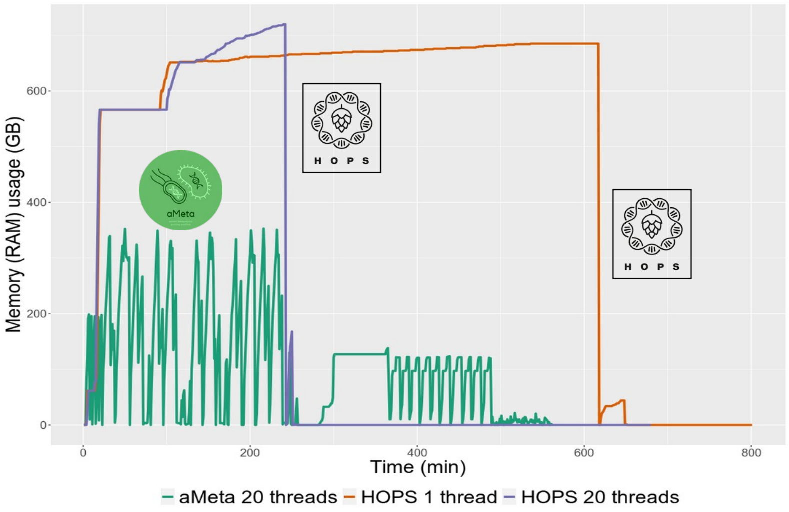
Authentication error



Detection error



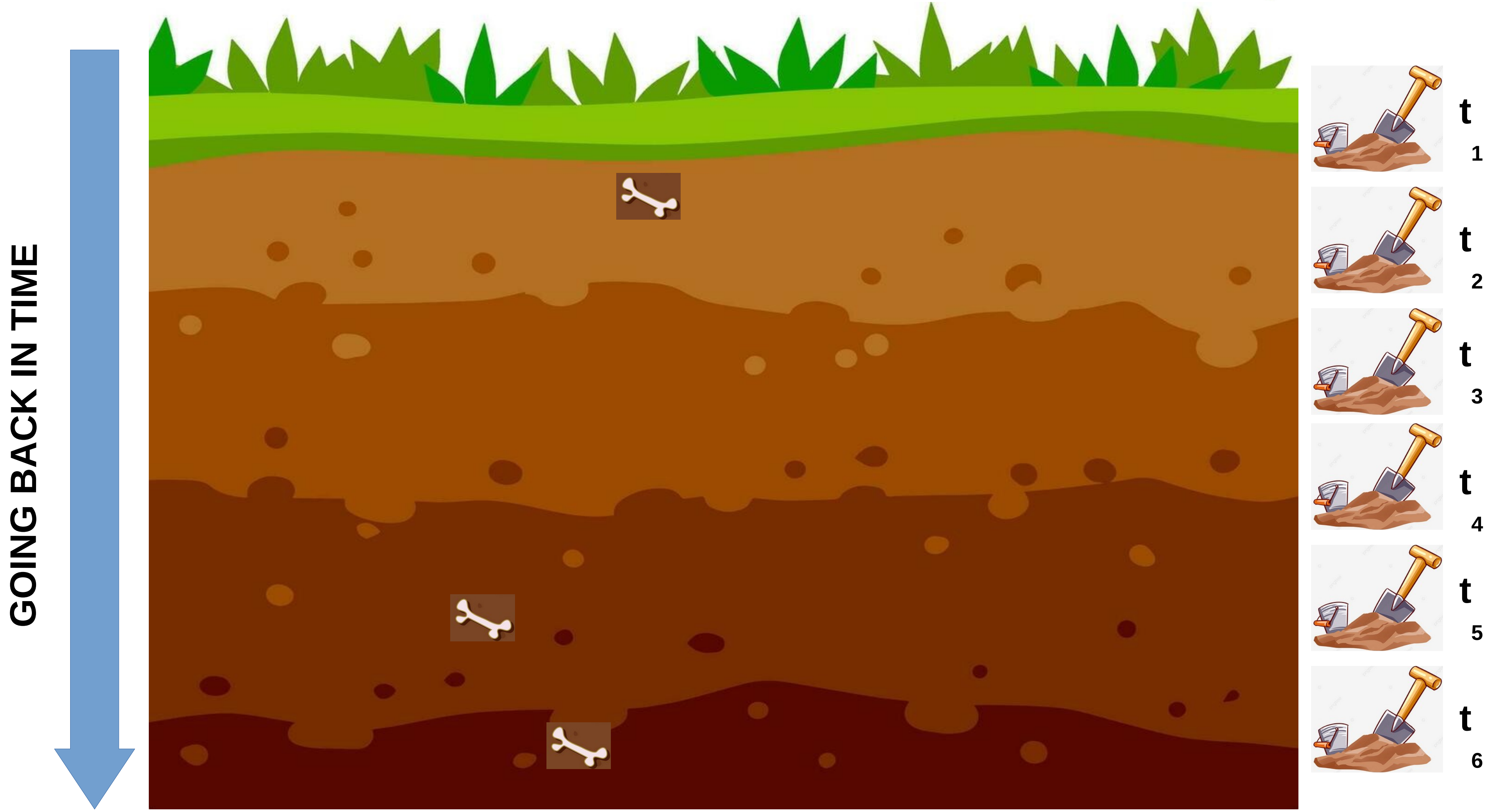
Resource usage



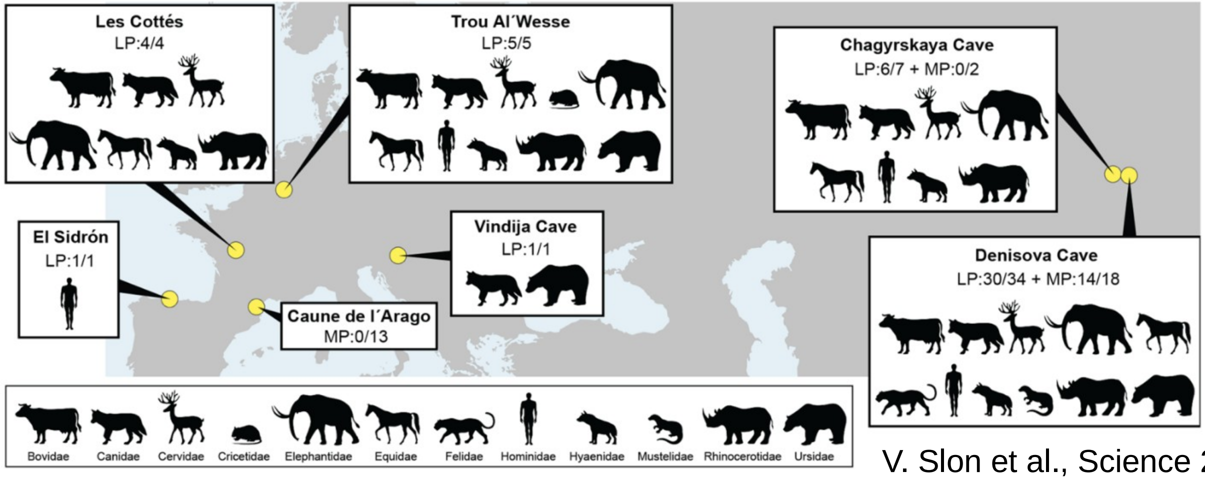
Pochon et al.
Genome Biology 2023

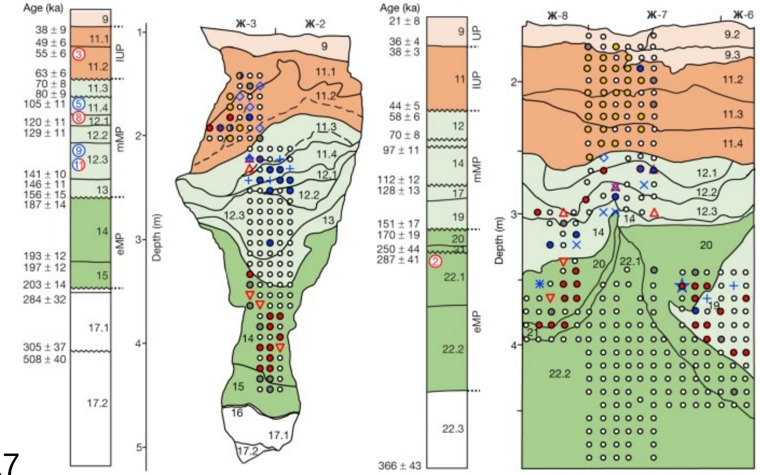
Computational challenges in ancient environmental metagenomics

Environmental and sedimentary aDNA



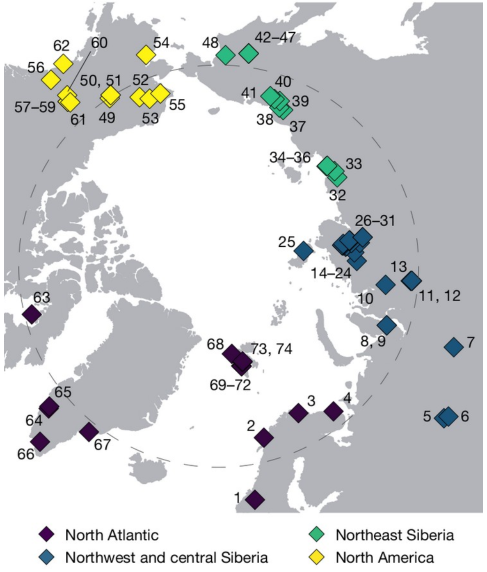
Environmental and sedimentary aDNA

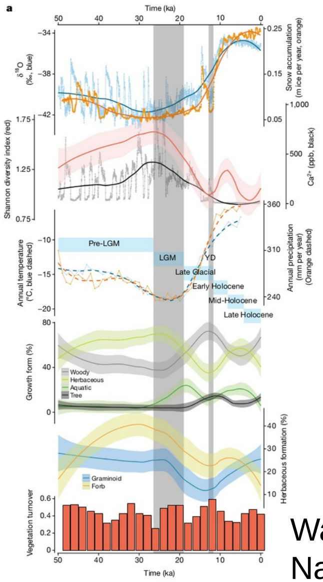




V. Slon et al., Science 2017

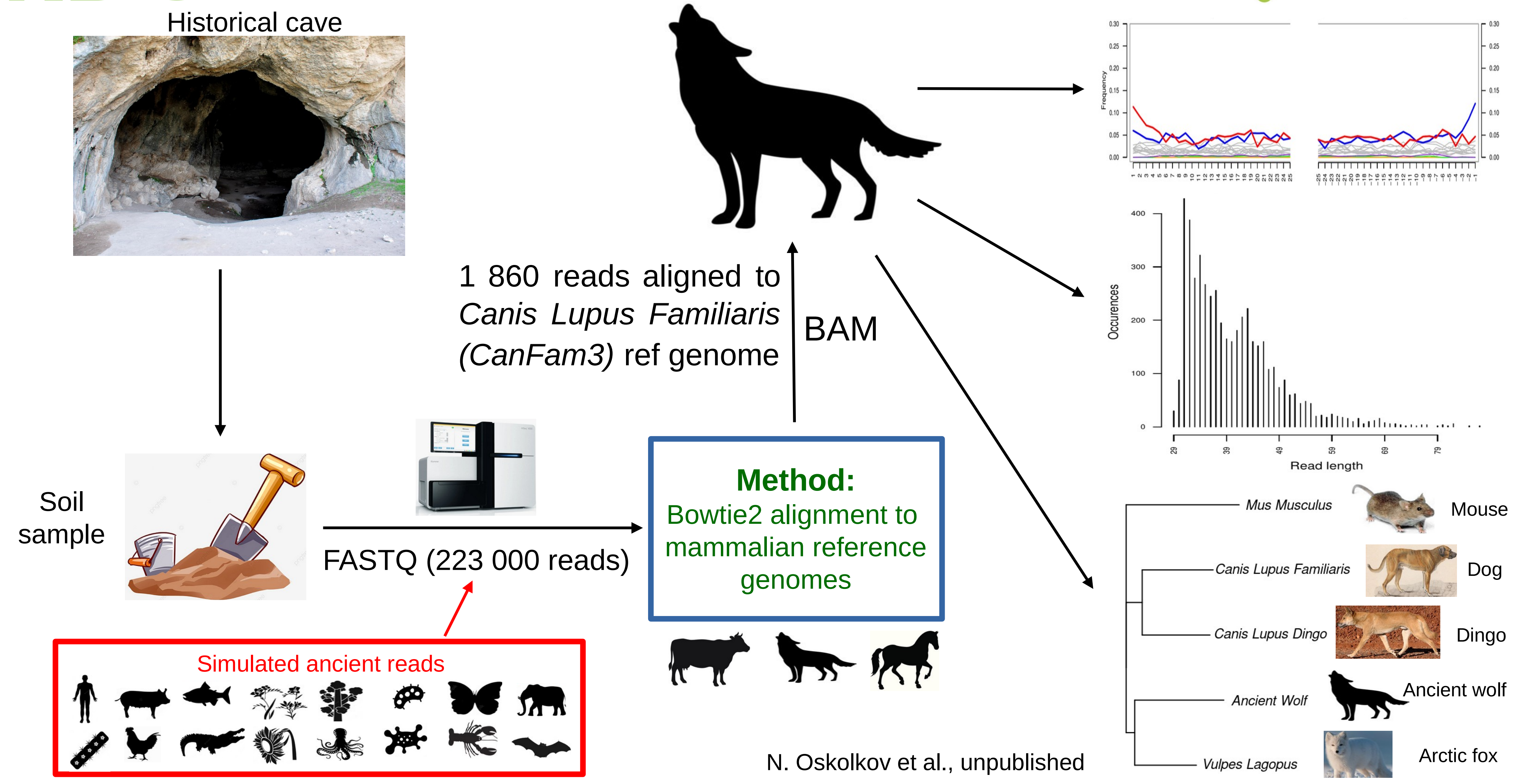
Jensen et al., Nat.Comm. 2017



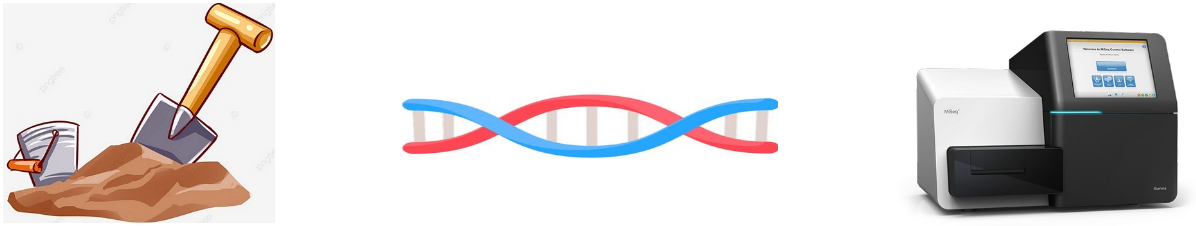


Wang et al., Nature 2021

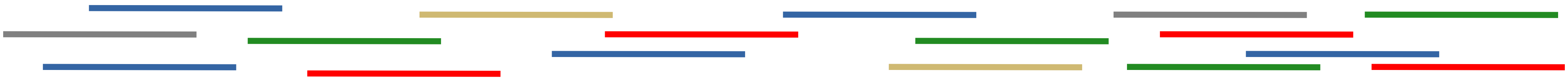
False-positives in environmental aDNA



Metagenomics requires large databases



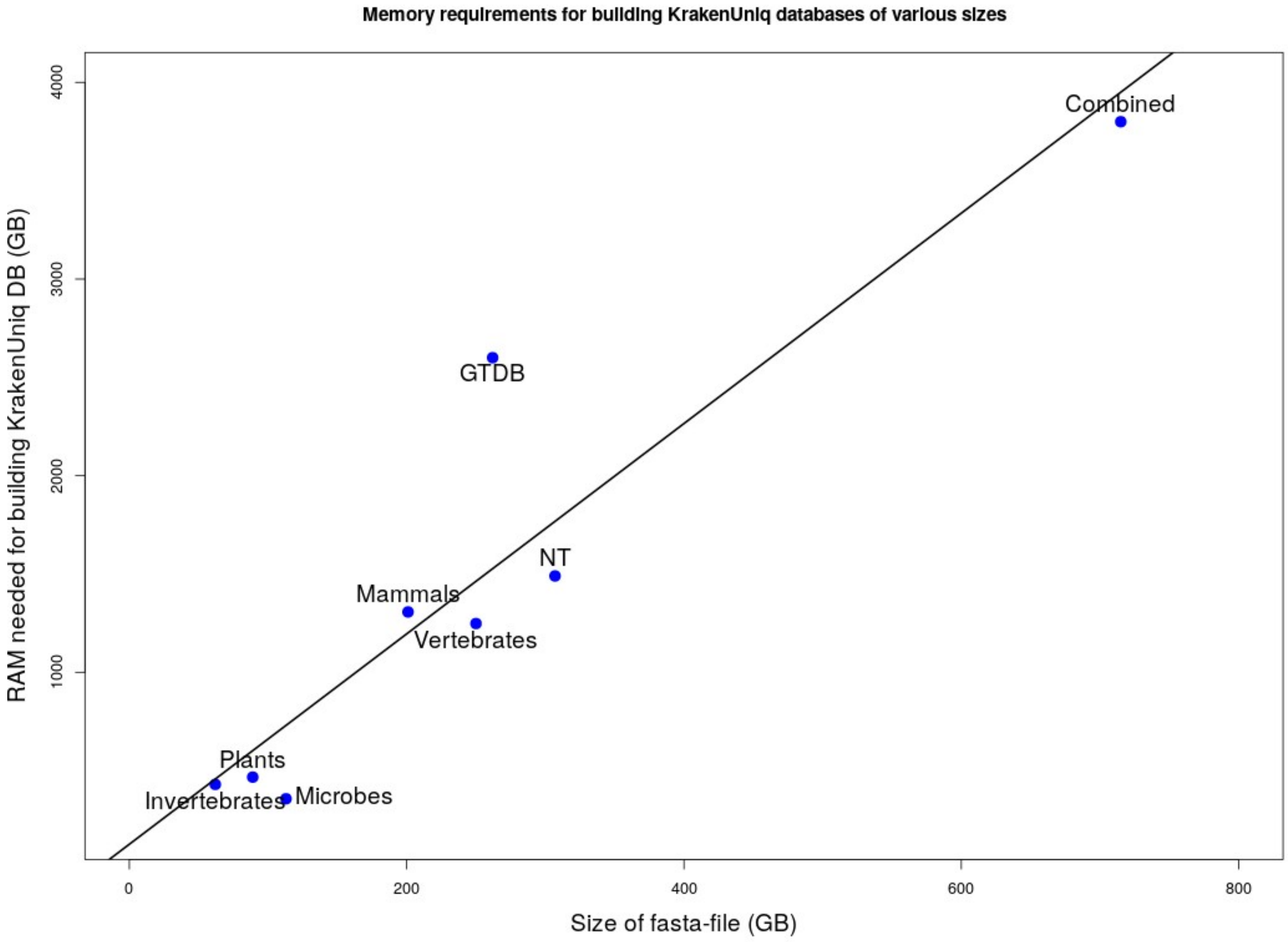
MIXTURE OF READS FROM DIFFERENT ORGANISMS



Mammals	Vertebrates	Plants	Invertebrates	Microbes

GRAND DATABASE

Large databases require a lot of RAM



GENEX: Oskolkov et al. GigaScience 2025

OXFORD

GIGA SCIENCE

GigaScience, 2025, 14, 1–14
DOI: 10.1093/gigascience/giaf108
Technical Note

Improving taxonomic inference from ancient environmental metagenomes by masking microbial-like regions in reference genomes

Nikolay Oskolkov[1,*], Chenyu Jin[2,3,4], Samantha López Clinton[2,3,4], Benjamin Guinet[2,3], Flore Wijnands[2,5], Ernst Johnson[2,5], Verena E. Kutschera[6], Cormac M. Kinsella[3,7], Peter D. Heintzman[2,5], and Tom van der Valk[2,3,8]

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Abstract

Ancient environmental DNA is increasingly vital for reconstructing past ecosystems, particularly when paleontological and archaeological tissue remains are absent. Detecting ancient plant and animal DNA in environmental samples relies on using extensive eukaryotic reference genome databases for profiling metagenomics data. However, many eukaryotic genomes contain regions with high sequence similarity to microbial DNA, which can lead to the misclassification of bacterial and archaeal reads as eukaryotic. This issue is especially problematic in ancient eDNA datasets, where plant and animal DNA is typically present at very low abundance. In this study, we present a method for identifying bacterial- and archaeal-like sequences in eukaryotic genomes and apply it to nearly 3,000 reference genomes from NCBI RefSeq and GenBank (vertebrates, invertebrates, plants) as well as the 1,323 PhyloNorway plant genome assemblies from herbarium material from northern high-latitude regions. We find that microbial-like regions are widespread across eukaryotic genomes and provide a comprehensive resource of their genomic coordinates and taxonomic annotations. This resource enables the masking of microbial-like regions during profiling analyses, thereby improving the reliability of ancient environmental metagenomic datasets for downstream analyses.

Keywords: environmental DNA, ancient metagenomics, microbial-like regions

Introduction

Ancient environmental DNA (aeDNA) is a tool for studying past ecosystems, especially in contexts where traditional archaeological and paleontological tissue remains, such as bones and seeds, are absent [1–4]. It consists of genetic traces left by organisms in the environment, such as soil, sediments, or ice, and allows for the reconstruction of past biodiversity and ecological communities to provide insight into species extinction, vegetation changes, and ecosystem responses to climatic shifts and anthropogenic impacts.

to genomic reference databases. Consequently, the quality of both the aeDNA data and the reference databases is crucial for reliable inferences. Microbial-like sequences in reference genomic databases, originating either from nonendogenous sources (contamination) or from evolutionary similarity to microbial genomes (e.g., due to ancient horizontal gene transfer or the endosymbiotic origins of plastids), can be a potential source of false-positive taxonomic identifications. In such cases, microbial sequences present in aeDNA data may be mistakenly classified as belonging to a eukaryotic reference genome due to sequence similarity.

NikolayOskolkov Update README.md · e45859c · 19 hours ago · 48 Commits

data	modified nextflow pipeline	2 months ago
images	Add files via upload	19 hours ago
GTDB_fna2name.txt	added workflow files	7 months ago
GTDB_sliced_seqs_sliding_window.fna.gz	added workflow files	7 months ago
LICENSE.txt	Add files via upload	7 months ago
README.md	Update README.md	19 hours ago
detect_exogenous.sh	modified nextflow pipeline	2 months ago
environment.yaml	added nextflow framework	2 months ago
extract_coords.R	modified nextflow pipeline	2 months ago
extract_coords_micr_contam.R	major modification of codes	2 months ago
human_sliced_seqs_sliding_window.fna.gz	modified nextflow pipeline	2 months ago
main.nf	modified nextflow pipeline	2 months ago
micr_cont_detect.sh	major modification of codes	2 months ago
nextflow.config	major modification of codes	2 months ago
vignette.html	modified vignette	2 months ago
vignette.ipynb	modified vignette	2 months ago

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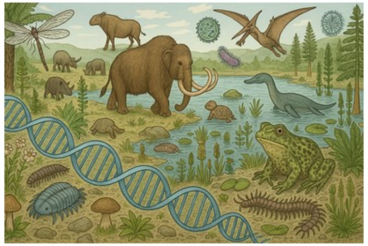
GENEX
GENOME EXOGENOUS



GENome EXogenous (GENEX) sequence detection

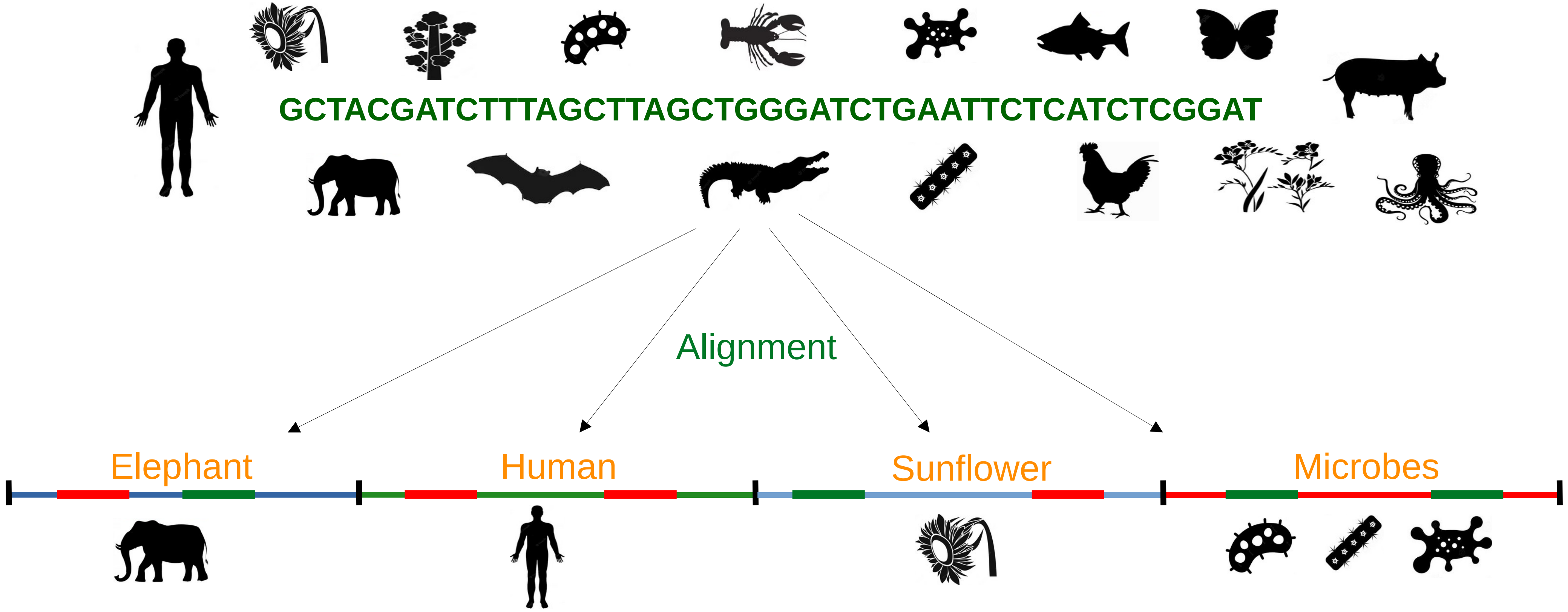
This is a computational workflow for detecting coordinates of microbial-like or human-like sequences in eukaryotic and procaryotic reference genomes. The workflow accepts a reference genome in FASTA-format and outputs coordinates of microbial-like (human-like) regions in BED-format. The workflow builds a Bowtie2 index of the reference genome and aligns pre-computed microbial (GTDB v.214 or NCBI RefSeq release 213) or

https://github.com/NikolayOskolkov/MCWorkflow

Uncertainty with sequence identification

Is it DNA or RNA?

Does it belong to the organism I am thinking about, or is it contamination?



Reference genomes: are they OK or contaminated?

Contamination within reference genomes

Research

Human contamination in bacterial genomes has created thousands of spurious proteins

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Contaminant sequences that appear in published genomes can cause numerous problems for downstream analyses, particularly for evolutionary studies and metagenomics projects. Our large-scale scan of complete and draft bacterial and archaeal genomes in the NCBI RefSeq database reveals that 2250 genomes are contaminated by human sequence. The contaminant sequences derive primarily from high-copy human repeat regions, which themselves are not adequately represented in the current human reference genome, GRCh38. The absence of the sequences from the human assembly offers a likely explanation for their presence in bacterial assemblies. In some cases, the contaminating contigs have been erroneously annotated as containing protein-coding sequences, which over time have propagated to create spurious protein "families" across multiple prokaryotic and eukaryotic genomes. As a result, 3437 spurious protein entries are currently present in the widely used nr and TrEMBL protein databases. We report here an extensive list of contaminant sequences in bacterial genome assemblies and the proteins associated with them. We found that nearly all contaminants occurred in small contigs in draft genomes, which suggests that filtering out small contigs from draft genome assemblies may mitigate the issue of contamination while still keeping nearly all of the genuine genomic sequences.

OPEN ACCESS Freely available online

PLoS one

Abundant Human DNA Contamination Identified in Non-Primate Genome Databases

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Department of Molecular and Cell Biology, University of Connecticut, Storrs, Connecticut, United States of America

Abstract

During routine screens of the NCBI databases using human repetitive elements we discovered an unlikely level of nucleotide identity across a broad range of phyla. To ascertain whether databases containing DNA sequences, genome assemblies and trace archive reads were contaminated with human sequences, we performed an in depth search for sequences of human origin in non-human species. Using a primate specific SINE, AluY, we screened 2,749 non-primate public databases from NCBI, Ensembl, JGI, and UCSC and have found 492 to be contaminated with human sequence. These represent species ranging from bacteria (*B. cereus*) to plants (*Z. mays*) to fish (*D. rerio*) with examples found from most phyla. The identification of such extensive contamination of human sequence across databases and sequence types warrants caution among the sequencing community in future sequencing efforts, such as human re-sequencing. We discuss issues this may raise as well as present data that gives insight as to how this may be occurring.

Citation: Longo MS, O'Neill MJ, O'Neill RJ (2011) Abundant Human DNA Contamination Identified in Non-Primate Genome Databases. PLoS ONE 6(2): e16410. doi:10.1371/journal.pone.0016410

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Competing Interests: The authors have declared that no competing interests exist.

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Correspondence

Origin of land plants revisited in the light of sequence contamination and missing data

Simon Laurin-Lemay, Henner Brinkmann, Hervé Philippe ✉

EMBO reports

Science & Society

Here, there, and everywhere

From PCRs to next-generation sequencing technologies and sequence databases, DNA contaminants creep in from the most unlikely places

Karl Gruber

PNAS

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SEE COMMENTARY

No evidence for extensive horizontal gene transfer in the genome of the tardigrade *Hypsibius dujardini*

Georgios Koutsovoulos[a], Sujai Kumar[a], Dominik R. Laetsch[a,b], Lewis Stevens[a], Jennifer Daub[a], Claire Conlon[a], Habib Maroon[a], Fran Thomas[a], Aziz A. Aboobaker[c], and Mark Blaxter[a,1]

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Edited by W. Ford Doolittle, Dalhousie University, Halifax, Canada, and approved March 1, 2016 (received for review January 8, 2016)

Tardigrades are meiofaunal ecdysozoans that are key to understanding the origins of Arthropoda. Many species of Tardigrada can survive extreme conditions through cryptobiosis. In a recent paper [Boothby TC, et al. (2015) *Proc Natl Acad Sci USA* 112(52):15976–15981], the authors concluded that the tardigrade *Hypsibius dujardini* had an unprecedented proportion (17%) of genes originating through functional horizontal gene transfer (fHGT) and speculated that fHGT was likely formative in the evolution of cryptobiosis. We independently sequenced the genome of *H. dujardini*. As expected from whole-organism DNA sampling, our raw data contained reads from nontarget genomes. Filtering using metagenomics approaches generated a draft *H. dujardini* genome assembly of 135 Mb with superior assembly metrics to the previously published assembly. Additional microbial contamination likely remains. We found no support for extensive fHGT. Among 23,021 gene predictions we identified 0.2% strong candidates for fHGT from bacteria and 0.2% strong candidates for fHGT from nonmetazoan eukaryotes. Cross-comparison of assemblies showed that the overwhelming majority of HGT candidates in the Boothby et al. genome derived from contaminants. We conclude that fHGT into *H. dujardini* accounts for at most 1–2% of genes and that the proposal that one-sixth of tardigrade genes originate from functional HGT events is an artifact of undetected contamination.

tardigrade | blobtools | contamination | metagenomics | horizontal gene transfer

cryptobiotic (24), but serves as a useful comparator for good cryptobiotic species (9).

Animal genomes can accrete horizontally transferred DNA, especially from germ line-transmitted symbionts (25), but the majority of transfers are nonfunctional and subsequently evolve neutrally and can be characterized as dead-on-arrival horizontal gene transfer (doaHGT) (25–27). Functional horizontal gene transfer (fHGT) can bring to a recipient genome new biochemical capacities and contrasts with gradualist evolution of endogenous genes to new function. The bdelloid rotifers *Adineta vaga* (28) and *Adineta ricciae* (29) have high levels of fHGT (~8%), and this has been associated with both their survival as phylogenetically ancient asexuals and their ability to undergo cryptobiosis (28–32). Different kinds of evidence are required to support claims of doaHGT compared with fHGT. Both are supported by phylogenetic proof of foreignness, linkage to known host genome-resident genes, in situ proof of presence on nuclear chromosomes (33), Mendelian inheritance (34), and phylogenetic perdurance (presence in all, or many individuals of a species, and presence in related taxa). Functional integration of a foreign gene into an animal genome requires adaptation to the new transcriptional environment including acquisition of spliceosomal introns, acclimatization to host base composition and codon use bias, and evidence of active transcription (e.g., in mRNA sequencing data) (35, 36).

Another source of foreign sequence in genome assemblies is contamination, which is easy to generate and difficult to sepa-

GENETICS

PeerJ

Unexpected cross-species contamination in genome sequencing projects

Samier Merchant[1,2], Derrick E. Wood[1,3] and Steven L. Salzberg[1,3,4]

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ABSTRACT

The raw data from a genome sequencing project sometimes contains DNA from contaminating organisms, which may be introduced during sample collection or sequence preparation. In some instances, these contaminants remain in the sequence even after assembly and deposition of the genome into public databases. As a result, searches of these databases may yield erroneous and confusing results. We used efficient microbiome analysis software to scan the draft assembly of domestic cow, *Bos taurus*, and identify 173 small contigs that appeared to derive from microbial contaminants. In the course of verifying these findings, we discovered that one genome, *Neisseria gonorrhoeae* TCDC-NG08107, although putatively a complete genome, contained multiple sequences that actually derived from the cow and sheep genomes. Our findings illustrate the need to carefully validate findings of anomalous DNA that rely on comparisons to either draft or finished genomes.

Steinegger and Salzberg *Genome Biology* (2020) 21:115
https://doi.org/10.1186/s13059-020-02023-1

Genome Biology

METHOD Open Access

Terminating contamination: large-scale search identifies more than 2,000,000 contaminated entries in GenBank

Check for updates

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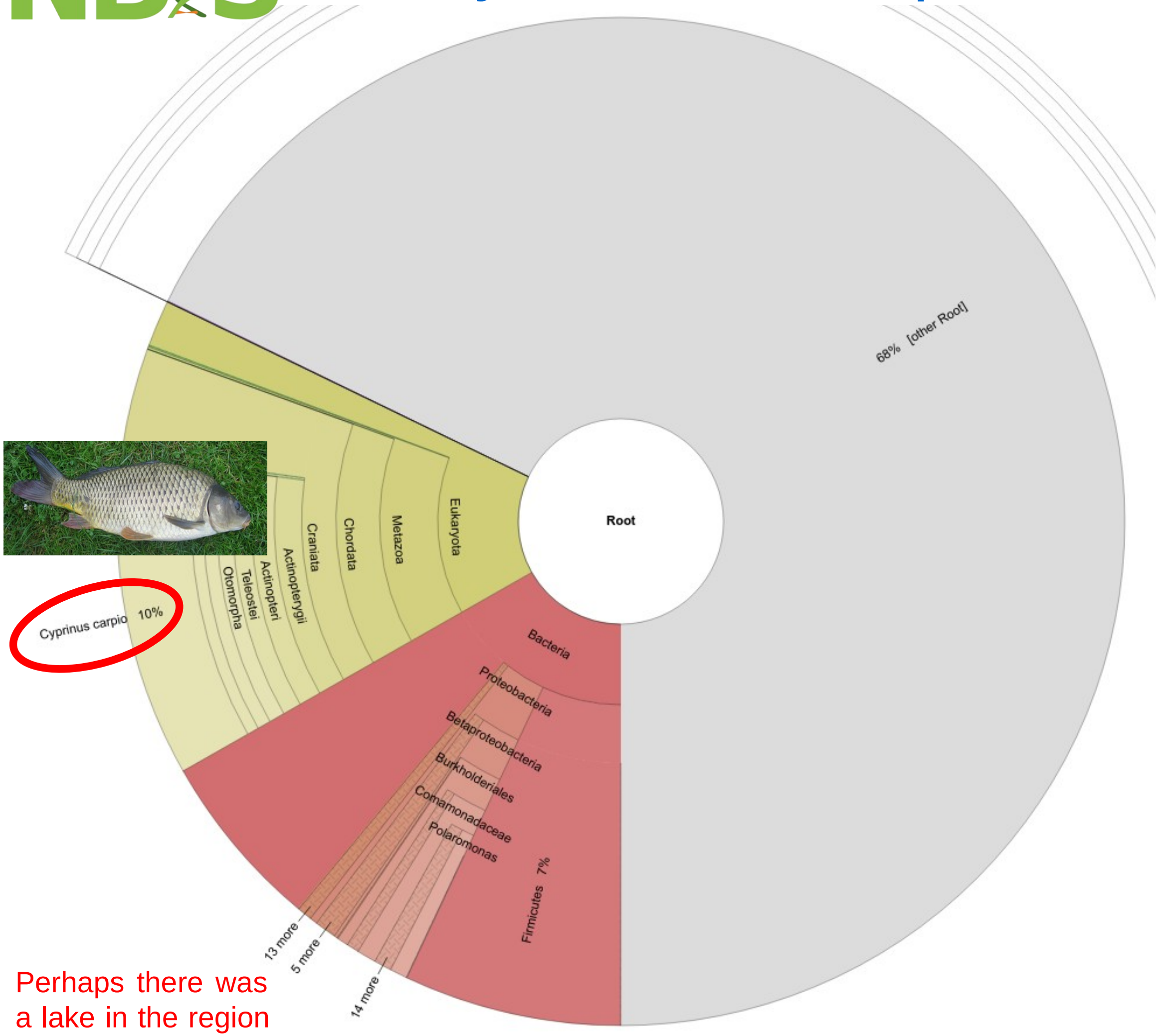
Full list of author information is available at the end of the article

Abstract

Genomic analyses are sensitive to contamination in public databases caused by incorrectly labeled reference sequences. Here, we describe Conterminator, an efficient method to detect and remove incorrectly labeled sequences by an exhaustive all-against-all sequence comparison. Our analysis reports contamination of 2,161,746, 114,035, and 14,148 sequences in the RefSeq, GenBank, and NR databases, respectively, spanning the whole range from draft to "complete" model organism genomes. Our method scales linearly with input size and can process 3.3 TB in 12 days on a 32-core computer. Conterminator can help ensure the quality of reference databases. Source code (GPLv3): https://github.com/martin-steinegger/conterminator

Keywords: Genomes, Contamination, Software, RefSeq, GenBank

My favorite: “carp in the soil” example



Perhaps there was a lake in the region where the soil sample came from?

No, you simply forgot to trim your adapters :)))

It turns out the Carp genome is full of Illumina adapters.

One of the first things we teach people in our NGS courses is how to remove adapters. It’s not hard – we use CutAdapt, but many other tools exist. It’s simple, but really important – with De Bruijn graphs you will get paths through the graphs converging on kmers from adapters; and with OLC assemblers you will get spurious overlaps. With gap-fillers, it’s possible to fill the gaps with sequences ending in adapters, and this may be what happened in the Carp genome.

Why then are we finding such elementary mistakes in such important papers? Why aren’t reviewers picking up on this? It’s frustrating.

This is a separate, but related issue, to genomic contamination – the Wheat genome has PhiX in it; tons of bacterial genomes do too; and lots of bacterial genes were problematically included in the Tardigrade genome and declared as horizontal gene transfer.

Bioinformatics Bits and Bobs

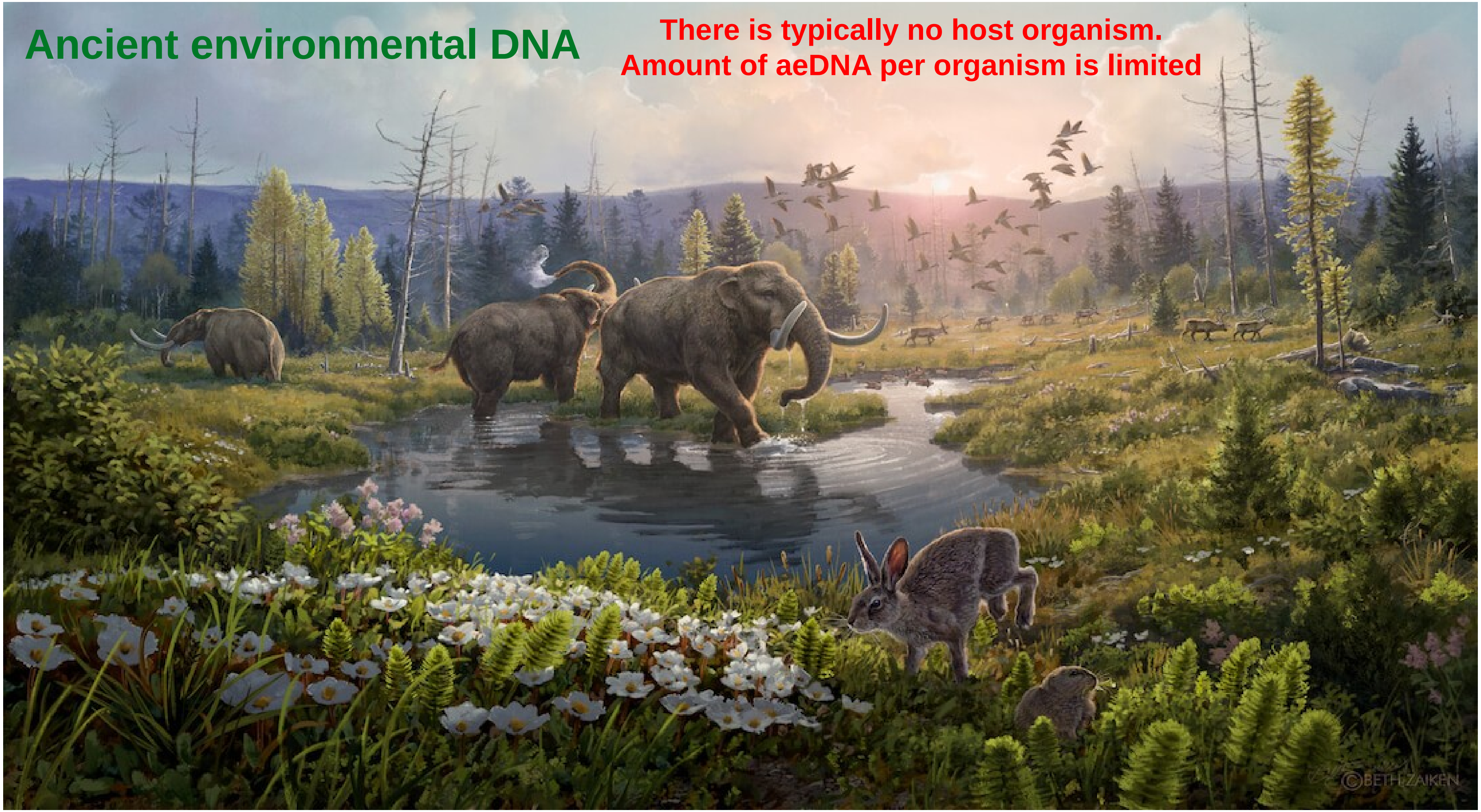
Rare and random blog posts about bioinformatics, genomics and evolution.

Monday, 29 September 2014

Why you should QC your reads AND your assembly

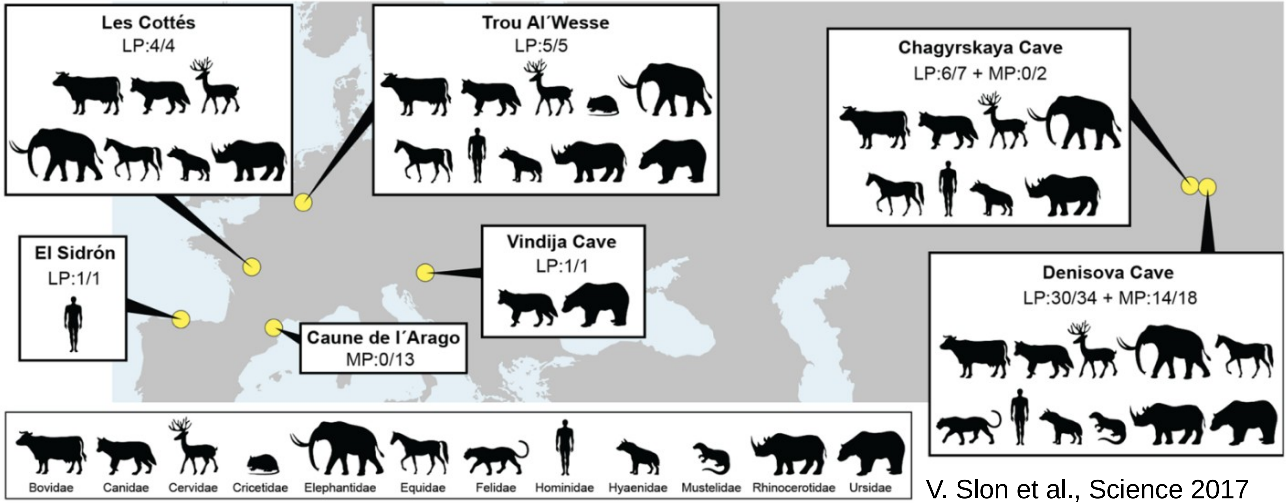
The genome sequence of the Common Carp *Cyprinus carpio* was published in Nature last week. By coincidence, I was doing some QC on some domesticated Ferret (*Mustela ptorius furo*) reads, which had thrown some kmer warnings in the FastQC tool. I blasted the kmers in NCBI and was quite perplexed by the number of hits that I found in the carp genome. Nearly all of the first 150 hits were all from the carp genome. Anyway, I looked a bit further into my odd kmers and it turns out that they were the ends of some Illumina adapter sequences that had presumably been incorporated into the paired-reads on the shorter ends of the insert size. This then took me back to the Carp Genome - what had creeped into that?

When reference genome quality is critical

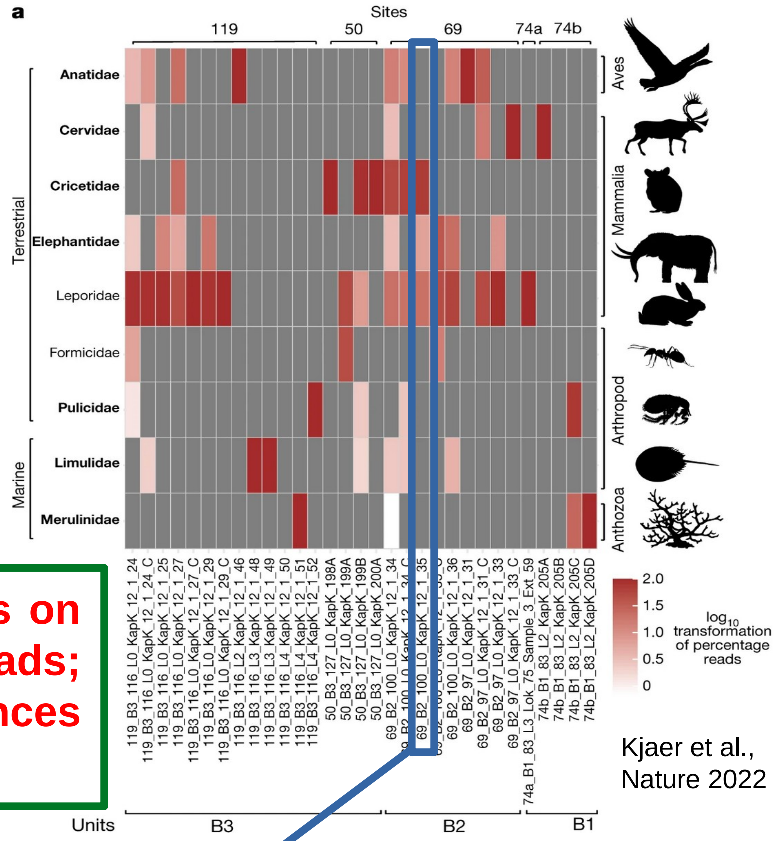


An artist's impression of the Kap København formation two-million years ago in a time where the temperature was significantly warmer than northernmost Greenland today. Artist: Beth Zaiken / bethzaiken.com

Environmental and sedimentary aDNA

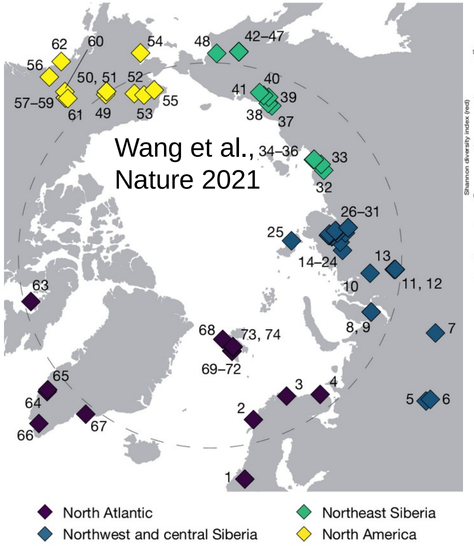


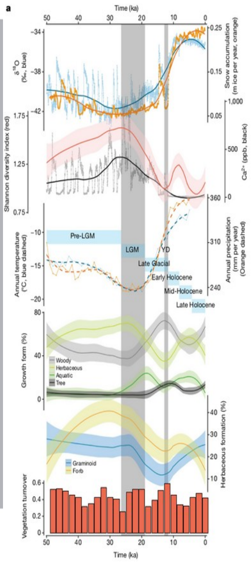
V. Slon et al., Science 2017



Kjaer et al., Nature 2022

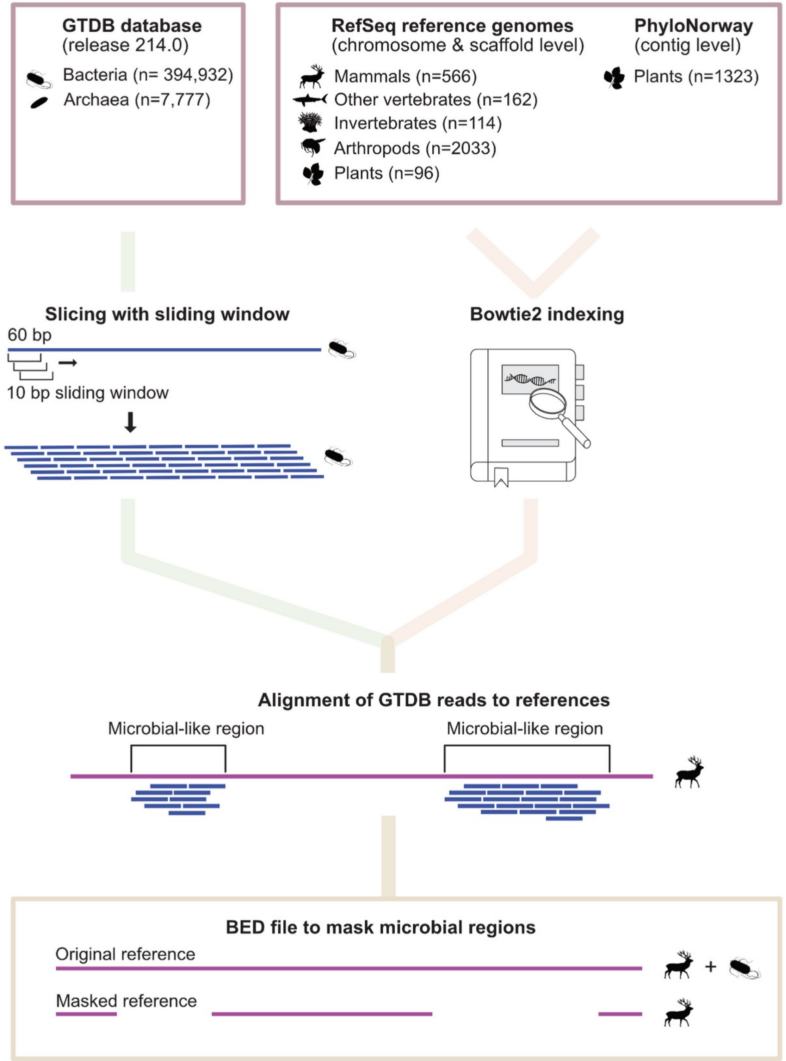
If organism detection relies on a very limited number of reads; hence, high-quality references are particularly important!

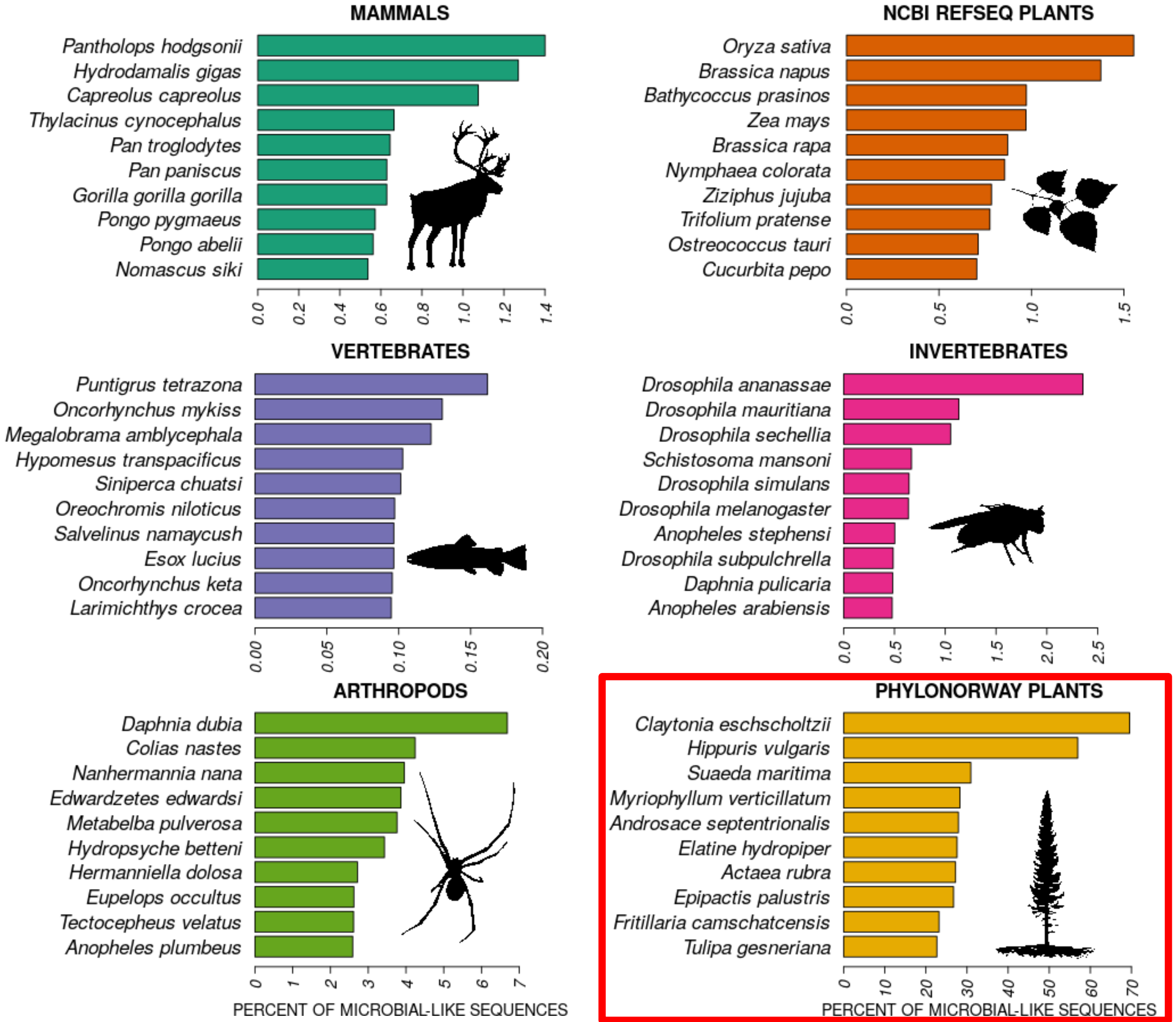




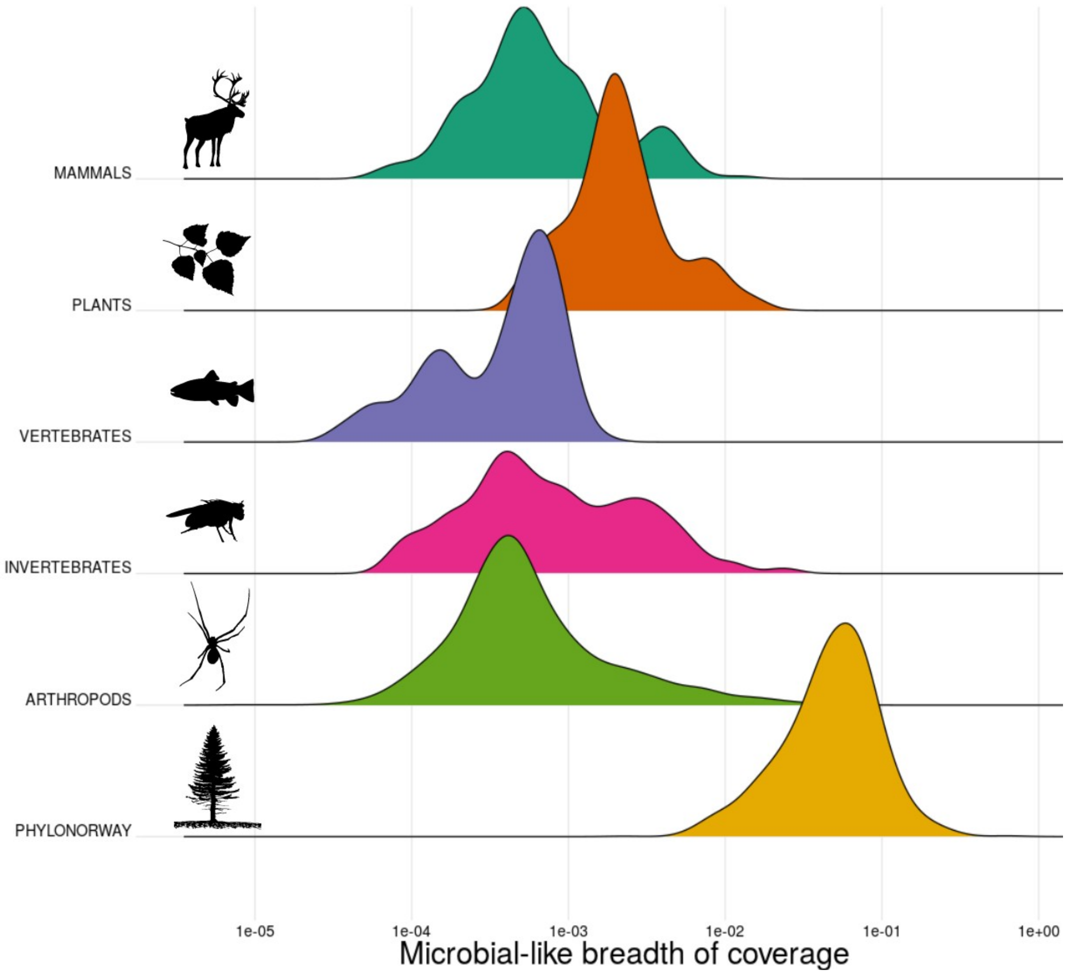
A	S	T	U	V	W
Taxa	69_B2_100_L0_KapK_12_1_34	69_B2_100_L0_KapK_12_1_34_C	69_B2_100_L0_KapK_12_1_35	9_B2_100_L0_KapK_12_1_35_C	69_B2_100_L0_KapK_12_1_36
Anatidae	372	3	0	0	70
Cervidae	84		0	0	0
Cricetidae	1331	17	1890	0	0
Elephantidae	85		171	48	118
Formicidae	0		0	19	0
Leporidae	542	5	549	61	313
Limulidae	66		0	0	23
Merulinidae	18		0	0	0
Pulicidae	0		0	0	0

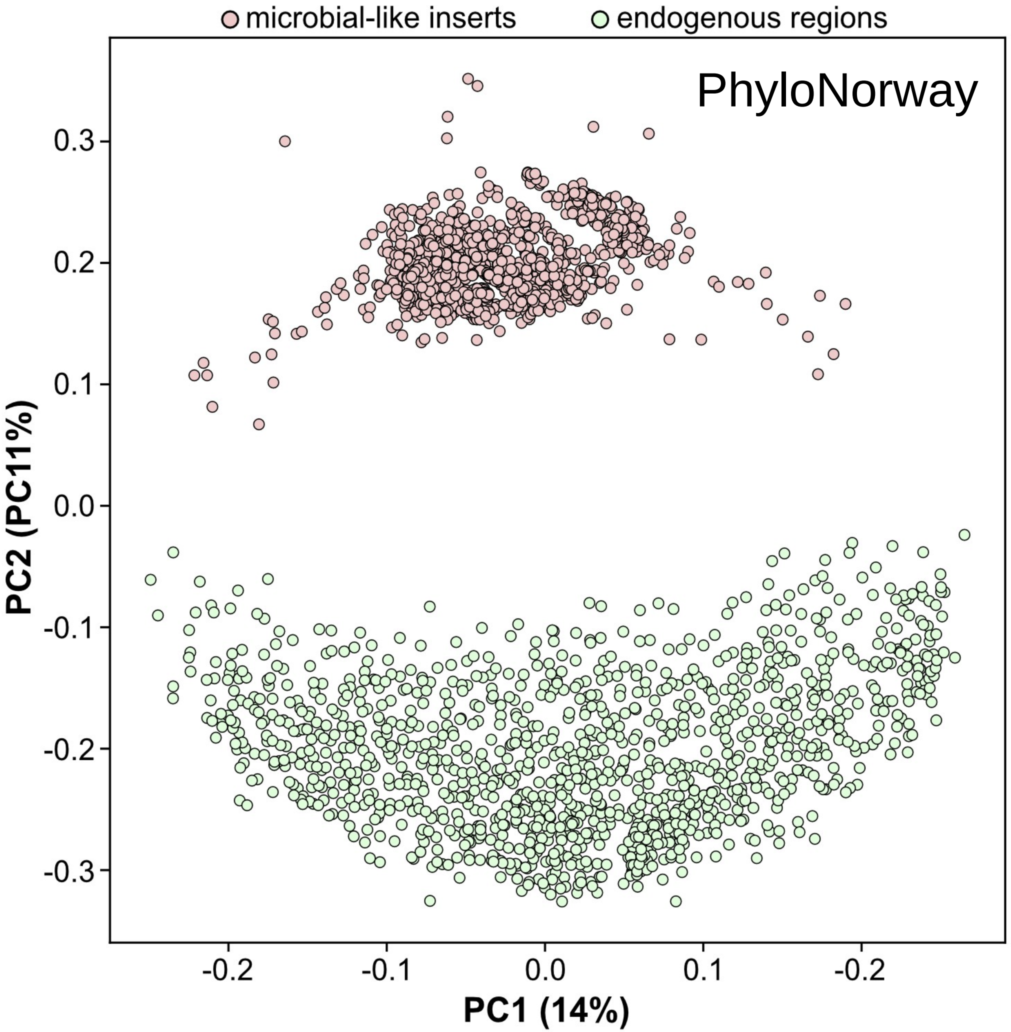
Microbial-like sequences in eukaryotes





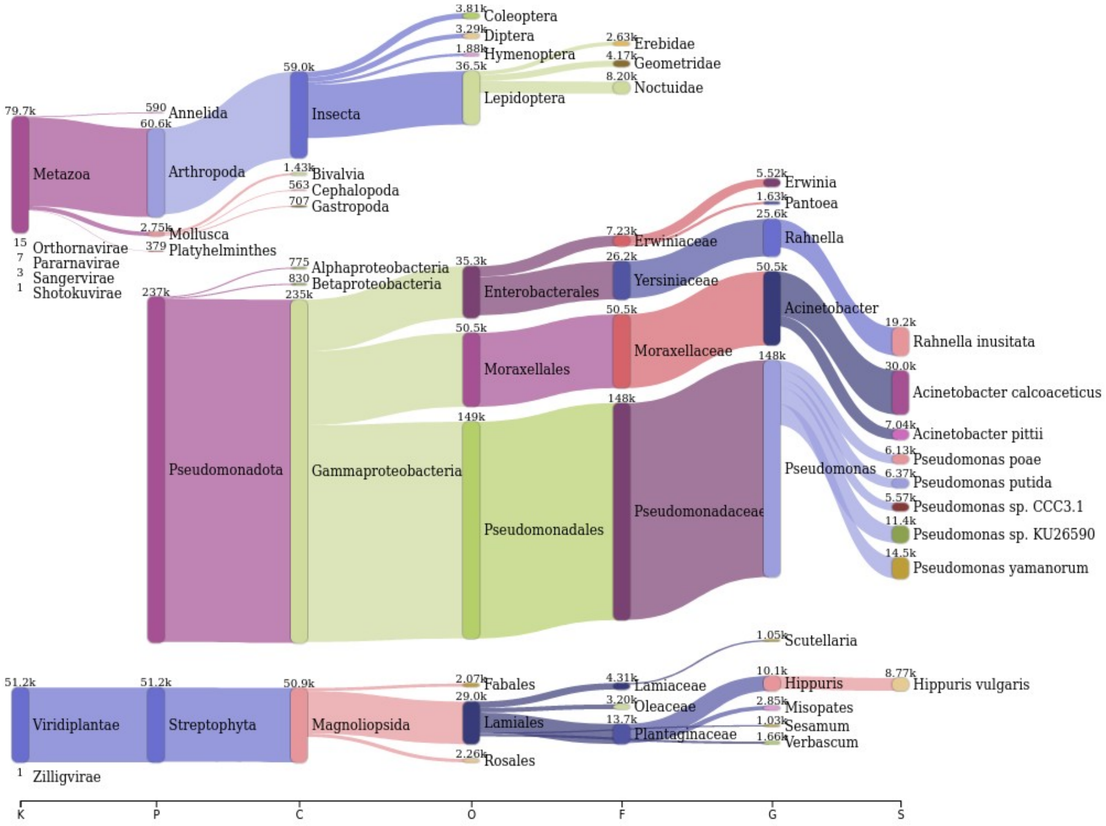
PhyloNorway dataset of Nordic plants

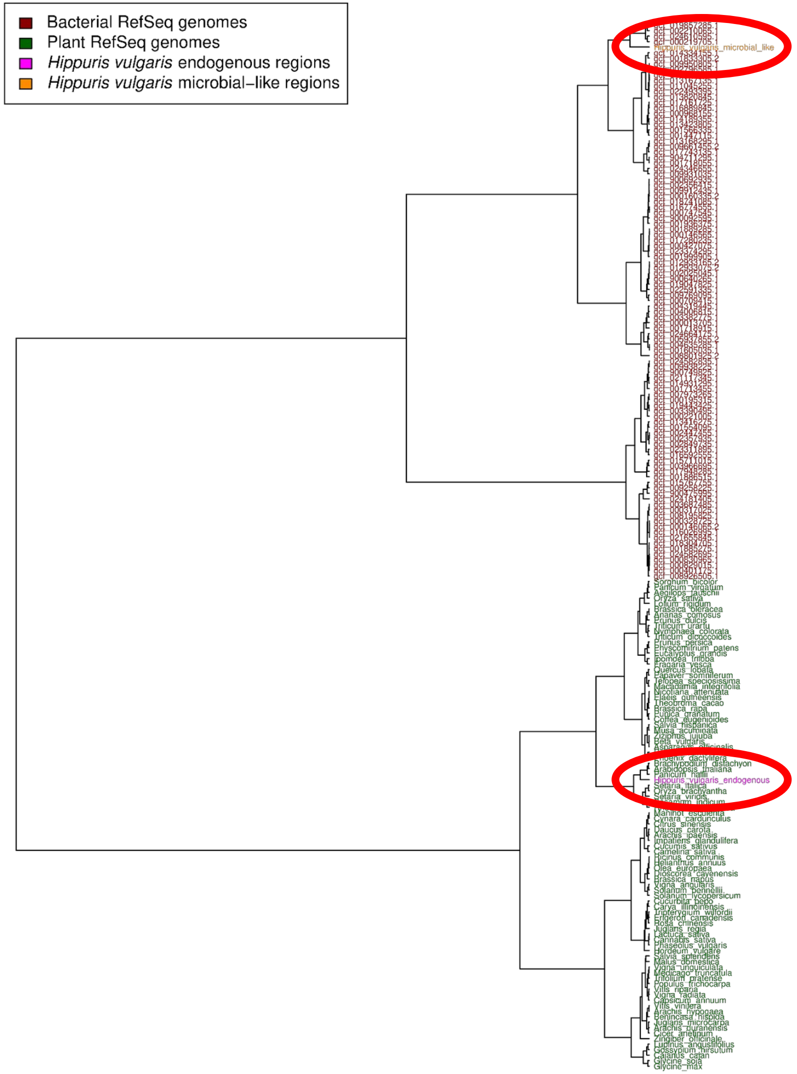




Followup *Hippuris vulgaris* from PhyloNorway

Screening *Hippuris vulgaris* contigs with Kraken2 against NCBI NT DB





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